



Rapport de TP :

Bases de données NoSQL

MongoDB : CRUD, Requêtes Avancées & Map-Reduce

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Introduction

L'objectif de ce TP est de se familiariser avec les bases de données orientées documents à travers le SGBD NoSQL MongoDB. L'environnement a été mis en place en démarrant un serveur mongod et un client interactif mongo. Nous avons ensuite manipulé une base locale avec des opérations CRUD, puis nous avons interrogé une base plus riche (gym) à travers des requêtes avancées et le paradigme Map-Reduce.

Exercice 1 : Installation, création d'une BD et opérations CRUD

1. Installation et lancement

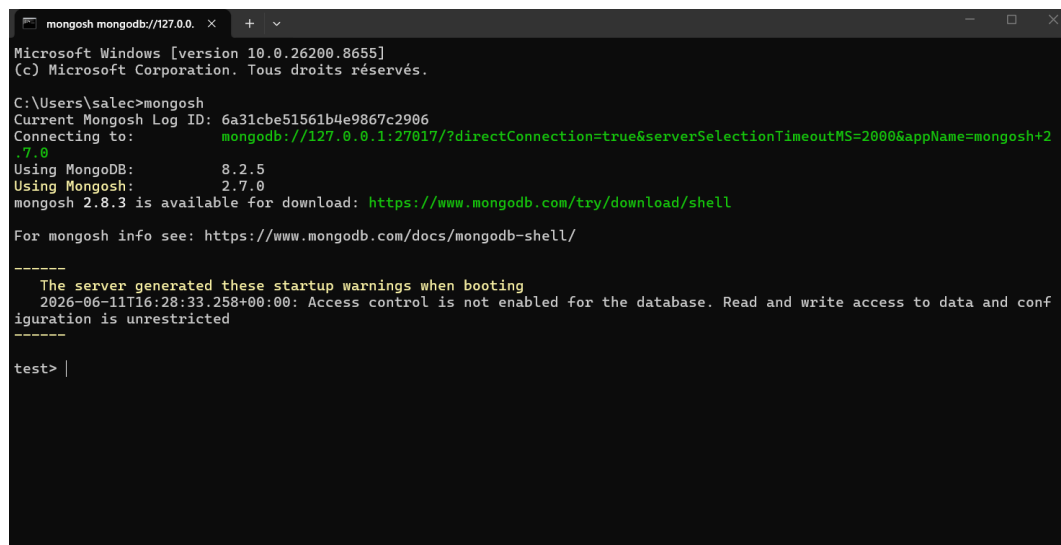
Après avoir téléchargé et configuré les répertoires de données (C:\data\db), le serveur a été lancé, suivi du client interactif.

2. Serveur mongo

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\salec\OneDrive\Bureau\BD_NoSQL\TP_BD-no-SQL>mkdir C:\data\db
C:\Users\salec\OneDrive\Bureau\BD_NoSQL\TP_BD-no-SQL>cd C:\data\db
C:\data\db>mongod
{"ts":{"sdate":"2026-06-16T22:18:30.186400000","s":"I","c":"-","id":8991200,"ctx":{"thread1","msg":"Shuffling initializers","attr":{"seed":"561033151"}}},"sdate":"2026-06-16T22:18:30.200400000","s":"I","c":"CONTROL","id":97374,"ctx":{"thread1","msg":"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"},"sdate":"2026-06-16T22:18:30.319400000","s":"I","c":"NETWORK","id":4915701,"ctx":{"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}},"sdate":"2026-06-16T22:18:30.320400000","s":"I","c":"CONTROL","id":5945603,"ctx":{"thread1","msg":"Multi threading initialized"},"sdate":"2026-06-16T22:18:30.321400000","s":"I","c":"CONTROL","id":4615611,"ctx":{"initandlisten","msg":"MongoDB starting","attr":{"pid":6584,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"SLK"}}},"sdate":"2026-06-16T22:18:30.321400000","s":"I","c":"CONTROL","id":23398,"ctx":{"initandlisten","msg":"Target operating system minimum version","attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}},"sdate":"2026-06-16T22:18:30.321400000","s":"I","c":"CONTROL","id":23403,"ctx":{"initandlisten","msg":"Build Info","attr":{"buildInfo":{"version":"8.2.5","gitversion":"4471a13894b34666c48a1f79451f2fa49f8f5df","modules":["allocator":"tcmalloc-gperf","environment":{"distmod":"windows"},"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"10.0 (build 26200)"},"modules":["allocator":"tcmalloc-gperf","environment":{"distmod":"windows"},"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"10.0 (build 26200)"}]]},"sdate":"2026-06-16T22:18:30.321400000","s":"I","c":"CONTROL","id":51765,"ctx":{"initandlisten","msg":"Operating System","attr":{"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"10.0 (build 26200)"},"modules":["allocator":"tcmalloc-gperf","environment":{"distmod":"windows"},"os":{"name":"Microsoft Windows Workstation (build 26200)","version":"10.0 (build 26200)"}]]},"sdate":"2026-06-16T22:18:30.322400000","s":"I","c":"CONTROL","id":21951,"ctx":{"initandlisten","msg":"Options set by command line","attr":{"options":{}}},"sdate":"2026-06-16T22:18:30.322400000","s":"I","c":"NETWORK","id":11621101,"ctx":{"initandlisten","msg":"Overriding max connections to honor 'capMemoryConsumptionForPreAuthBuffers' settings","attr":{"limit":"201044"}}},"sdate":"2026-06-16T22:18:30.323400000","s":"I","c":"NETWORK","id":1068601,"ctx":{"initandlisten","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":{"tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"}}},"sdate":"2026-06-16T22:18:30.329400000","s":"I","c":"STORAGE","id":22315,"ctx":{"initandlisten","msg":"Opening WiredTiger","attr":{"config":{"create,cache_size=734M,session_max=33000,eviction={threads_min=4,threads_max=4},config_base=false,statistics={fast},log=(enabled=true,remove=true,path=journal,compressor=snappy),builtin_extension_config={zstd={compression_level=0}},file_manager={close_idle_time=600,close_scan_interval=10,close_handle_minimum=2000},statistics_log=(wait=0),json_output={error,message},verbose={recovery_progress:1,checkpoint_progress:1,compact_progress:1,live_restore_progress:1,backup:0,checkpoint:0,compact:0,eviction:0,fileops:0,history_store:0,live_restore:0,recovery:0,rts:0,salvage:0,tiered:0,timestamp:0,transaction:0,verify:0,log:0},prefetch=(available=true,default=false),}}},"sdate":"2026-06-16T22:18:30.339400000","s":"I","c":"WTRECOV","id":22430,"ctx":{"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1781648310,"ts_usec":339423,"thread":"6584:140723736970784","session_name":"wiredtiger_open","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"opening the WiredTiger library"}}},"sdate":"2026-06-16T22:18:30.358400000","s":"I","c":"WTRECOV","id":22430,"ctx":{"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1781648310,"ts_usec":358429,"thread":"6584:140723736970784","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"connection configuration string parsing completed"}}},"sdate":"2026-06-16T22:18:30.396400000","s":"I","c":"WTRECOV","id":22430,"ctx":{"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1781648310,"ts_usec":396429,"thread":"6584:140723736970784","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting WiredTiger utility threads"}}},"sdate":"2026-06-16T22:18:30.419400000","s":"I","c":"WTRECOV","id":22430,"ctx":{"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1781648310,"ts_usec":419429,"thread":"6584:140723736970784","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting WiredTiger utility threads"}}},"sdate":"2026-06-16T22:18:30.420400000","s":"I","c":"WTRECOV","id":22430,"ctx":{"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1781648310,"ts_usec":419605,"thread":"6584:140723736970784","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting WiredTiger utility threads"}}}}
```

Figure 1: mongod

3. Client mongo



```
mongosh mongodb://127.0.0.1:27017
Microsoft Windows [version 10.0.26200.8655]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\salec>mongosh
Current Mongosh Log ID: 6a31cbe51561b4e9867c2906
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:      8.2.5
Using Mongosh:       2.7.0
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-06-11T16:28:33.258+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
  -----

test> |
```

Figure 2 : mongosh

5. Création d'une BD Mongo et opération CRUD

J'ai sélectionné la base info (ce qui la crée à la volée) et j'ai inséré trois documents dans la collection produits.

```
//question a
use info

//question b
db.produits.insert({nom: "Macbook Pro", fabricant: "Apple", prix: 11435.99, options: ["Intel Core i5", "Retina Display", "Long life battery"]})

//question c
db.produits.insert({nom: "Macbook Air", fabricant: "Apple", prix: 125794.73, ultrabook: true, options: ["Intel Core i7", "SSD", "Long life battery"]})

//question d
db.produits.insert({nom: "Thinkpad X230", fabricant: "Lenovo", prix: 114358.74, ultrabook: true, options: ["Intel Core i5", "SSD", "Long life battery"]})

//question e //Afficher tous les produits
db.produits.find().pretty()
```

```
mongosh mongodb://127.0.0.1:27017/
Using MongoDB: 8.2.5
Using Mongosh: 2.7.0
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----

info> db.produits.insertOne({
  nom: "Macbook Pro",
  fabriquant: "Apple",
  prix: 11435.99,
  options: ["Intel Core i5", "Retina Display", "Long life battery"]
})
{
  acknowledged: true,
  insertedId: ObjectId('6a31ceb31561b4e9867c2908')
}
info> show collections
produits
info> db.produits.find().pretty()
[
  {
    _id: ObjectId('6a31ceb31561b4e9867c2908'),
    nom: 'Macbook Pro',
    fabriquant: 'Apple',
    prix: 11435.99,
    options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
  }
]
```

Figure 3: exo1_question-5-a&b

```
mongosh mongodb://127.0.0.1:27017/
info> db.produits.insert({nom: "Macbook Air", fabriquant: "Apple", prix: 125794.73, ultrabook: true, options: ["Intel Core i7", "SSD", "Long life battery"]})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('6a31d0621561b4e9867c290a') }
}
info> db.produits.find().pretty()
[
  {
    _id: ObjectId('6a31ceb31561b4e9867c2908'),
    nom: 'Macbook Pro',
    fabriquant: 'Apple',
    prix: 11435.99,
    options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0621561b4e9867c290a'),
    nom: 'Macbook Air',
    fabriquant: 'Apple',
    prix: 125794.73,
    ultrabook: true,
    options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
  }
]
info> |
```

Figure 4: exo1-question-5-c

```
mongosh mongod://127.0.0.1
info> db.produits.insertOne({nom: "Thinkpad X230", fabricant: "Lenovo", prix: 114358.74, ultrabook: true, options: ["Intel Core i5", "SSD", "Long life battery"]})
{
  acknowledged: true,
  insertedId: ObjectId('6a31d0f21561b4e9867c290b')
}
info> db.produits.find().pretty()
[
  {
    _id: ObjectId('6a31ceb31561b4e9867c2908'),
    nom: 'Macbook Pro',
    fabricant: 'Apple',
    prix: 11435.99,
    options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0621561b4e9867c290a'),
    nom: 'Macbook Air',
    fabricant: 'Apple',
    prix: 125794.73,
    ultrabook: true,
    options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0f21561b4e9867c290b'),
    nom: 'Thinkpad X230',
    fabricant: 'Lenovo',
    prix: 114358.74,
    ultrabook: true,
  }
]
```

Figure 5: exo1_question5-d

```
mongosh mongod://127.0.0.1
info> db.produits.find().pretty()
[
  {
    _id: ObjectId('6a31ceb31561b4e9867c2908'),
    nom: 'Macbook Pro',
    fabricant: 'Apple',
    prix: 11435.99,
    options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0621561b4e9867c290a'),
    nom: 'Macbook Air',
    fabricant: 'Apple',
    prix: 125794.73,
    ultrabook: true,
    options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0f21561b4e9867c290b'),
    nom: 'Thinkpad X230',
    fabricant: 'Lenovo',
    prix: 114358.74,
    ultrabook: true,
    options: [ 'Intel Core i5', 'SSD', 'Long life battery' ]
  }
]
info>
```

Figure 6: exo1-question-5-e

Récupérer le premier produit :

```
db.produits.findOne()
```

```
mongosh mongodb://127.0.0.1 x + v
info> db.produits.findOne()
{
  _id: ObjectId('6a31ceb31561b4e9867c2908'),
  nom: 'Macbook Pro',
  fabricant: 'Apple',
  prix: 11435.99,
  options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
}
info> |
```

Figure 7: question-5-e1

Trouver l'id du Thinkpad et le récupérer avec son id :

```
var myId = db.produits.findOne({ nom: "Thinkpad X230" })._id;
db.produits.find({ _id: myId });
```

```
mongosh mongodb://127.0.0.1 x + v
info> var myId = db.produits.findOne({ nom: "Thinkpad X230" })._id;
info> db.produits.find({ _id: myId });
[
  {
    _id: ObjectId('6a31d0f21561b4e9867c290b'),
    nom: 'Thinkpad X230',
    fabricant: 'Lenovo',
    prix: 114358.74,
    ultrabook: true,
    options: [ 'Intel Core i5', 'SSD', 'Long life battery' ]
  }
]
info>
```

Figure 8: exo1-question5-e-2

Récupérer les produits dont le prix est supérieur à 13723 DA :

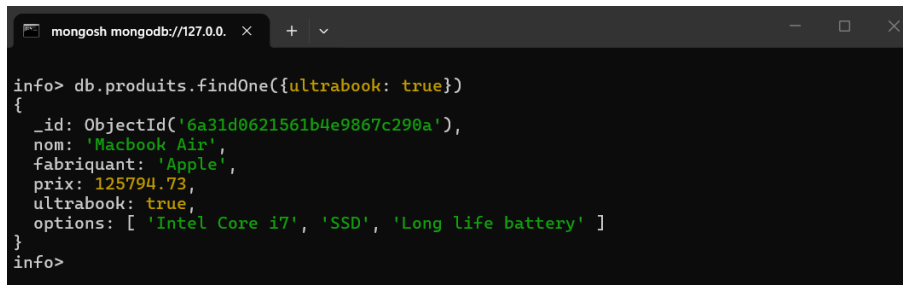
```
db.produits.find({prix: {$gt: 13723}})
```

```
mongosh mongodb://127.0.0.1 x + v
info> db.produits.find({prix: {$gt: 13723}})
[
  {
    _id: ObjectId('6a31d0621561b4e9867c290a'),
    nom: 'Macbook Air',
    fabricant: 'Apple',
    prix: 125794.73,
    ultrabook: true,
    options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0f21561b4e9867c290b'),
    nom: 'Thinkpad X230',
    fabricant: 'Lenovo',
    prix: 114358.74,
    ultrabook: true,
    options: [ 'Intel Core i5', 'SSD', 'Long life battery' ]
  }
]
info>
```

Figure 9: exo1_quest_5-e3

Récupérer le premier produit ayant le champ ultrabook à true :

```
db.produits.findOne({ultrabook: true})
```

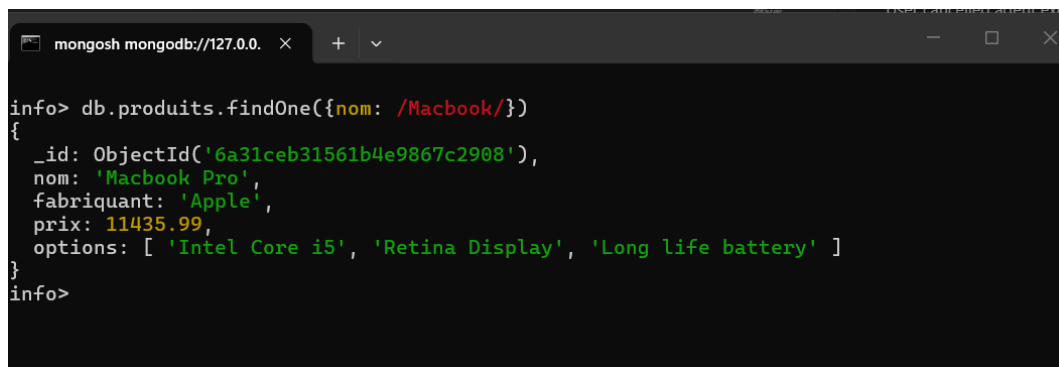


```
mongosh mongodb://127.0.0.1:27020/
info> db.produits.findOne({ultrabook: true})
{
  _id: ObjectId('6a31d0621561b4e9867c290a'),
  nom: 'Macbook Air',
  fabricant: 'Apple',
  prix: 125794.73,
  ultrabook: true,
  options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
}
info>
```

Figure 10: exo1_quest5-e-4

Récupérer le premier produit dont le nom contient Macbook :

```
db.produits.findOne({nom: /Macbook/})
```

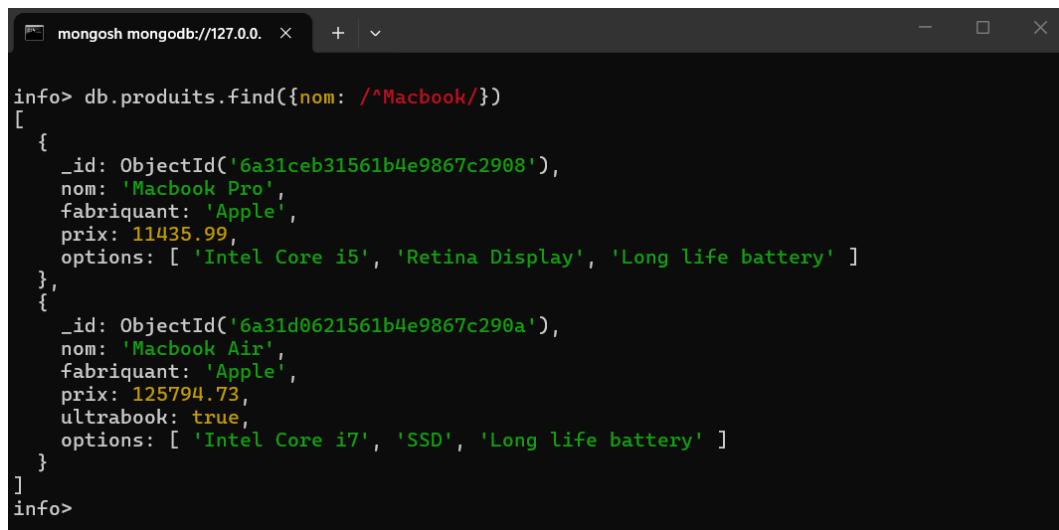


```
mongosh mongodb://127.0.0.1:27020/
info> db.produits.findOne({nom: /Macbook/})
{
  _id: ObjectId('6a31ceb31561b4e9867c2908'),
  nom: 'Macbook Pro',
  fabricant: 'Apple',
  prix: 11435.99,
  options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
}
info>
```

Figure 11: exo1_quest5-e5

Récupérer les produits dont le nom commence par Macbook :

```
db.produits.find({nom: /^Macbook/})
```



```
mongosh mongodb://127.0.0.1:27020/
info> db.produits.find({nom: /^Macbook/})
[
  {
    _id: ObjectId('6a31ceb31561b4e9867c2908'),
    nom: 'Macbook Pro',
    fabricant: 'Apple',
    prix: 11435.99,
    options: [ 'Intel Core i5', 'Retina Display', 'Long life battery' ]
  },
  {
    _id: ObjectId('6a31d0621561b4e9867c290a'),
    nom: 'Macbook Air',
    fabricant: 'Apple',
    prix: 125794.73,
    ultrabook: true,
    options: [ 'Intel Core i7', 'SSD', 'Long life battery' ]
  }
]
info>
```

Figure 12: exo1_quest5-e6

Supprimer les deux produits dont le fabricant est Apple :

```
db.produits.deleteMany({fabricant: "Apple"})
```

Supprimer le Lenovo X230 en utilisant uniquement son id :

```
var idLenovo = db.produits.findOne({nom: "Thinkpad X230"})._id;
db.produits.deleteOne({_id: idLenovo});
```

```

mongosh mongodb://127.0.0.1:27020/
info> var idLenovo = db.produits.findOne({nom: "Thinkpad X230"})._id;
info> db.produits.deleteOne({_id: idLenovo});
{ acknowledged: true, deletedCount: 1 }
info> db.produits.deleteMany({fabriquant: "Apple"})
{ acknowledged: true, deletedCount: 2 }
info>

```

Figure 13-14: exo1_quest5-e7

Exercice 2 : Requêtes avancées sur la base "gym" et Map-Reduce

Après restauration de la base via la commande de terminal mongorestore -d gym C:\MongoDB\gym, nous pouvons effectuer des interrogations ciblées.

3. Requêtes d'interrogation

a. Quels sont les sportifs (identifiant, nom et prénom) qui ont entre 20 et 30 ans ?

```

db.getCollection('Sportifs').find(
  { Age: { $gte: 20, $lte: 30 } },
  { IdSportif: 1, Nom: 1, Prenom: 1, _id: 0 }
)

```

Résultat :

```
```javascript
```

```

{"IdSportif": 1, "Nom": "BOUTAHAR", "Prenom": "Abderahim" }
{"IdSportif": 2, "Nom": "KERVADDEC", "Prenom": "Yann" }
{"IdSportif": 3, "Nom": "HUE", "Prenom": "Pascale" }
{"IdSportif": 5, "Nom": "COMES", "Prenom": "Sylvie" }
{"IdSportif": 6, "Nom": "RETALDI", "Prenom": "Sophie" }
{"IdSportif": 7, "Nom": "GOMEZ", "Prenom": "Diego" }
{"IdSportif": 9, "Nom": "ANTUNES", "Prenom": "Gaëlle" }
{"IdSportif": 10, "Nom": "BLANDET", "Prenom": "Arnaud" }
{"IdSportif": 11, "Nom": "BONNET", "Prenom": "Christophe" }
{"IdSportif": 12, "Nom": "BORREL", "Prenom": "Benoît" }
{"IdSportif": 13, "Nom": "BOYON", "Prenom": "Julien" }
{"IdSportif": 14, "Nom": "CAILLIOT", "Prenom": "Stéphanie" }
{"IdSportif": 15, "Nom": "CHAMPENOIS", "Prenom": "Cédric" }
{"IdSportif": 16, "Nom": "COLAS", "Prenom": "Michaël" }
{"IdSportif": 17, "Nom": "COLOMB", "Prenom": "Michaël" }
{"IdSportif": 18, "Nom": "HOSNI", "Prenom": "Leila" }
{"IdSportif": 19, "Nom": "TESTEMONT", "Prenom": "Laurent" }
{"IdSportif": 20, "Nom": "TIENER", "Prenom": "Thomas" }
{"IdSportif": 21, "Nom": "LE BOUCHER", "Prenom": "Denis" }
{"IdSportif": 22, "Nom": "LEDUFAUD", "Prenom": "Pierre" }
{"IdSportif": 23, "Nom": "LEGRAND", "Prenom": "David" }
{"IdSportif": 24, "Nom": "LEJEUNE", "Prenom": "Sylvaine" }
{"IdSportif": 25, "Nom": "MARTORA", "Prenom": "Fabrice" }

```



```
{ "IdSportif" : 26, "Nom" : "MICHEL", "Prenom" : "Frédérique" }
{ "IdSportif" : 27, "Nom" : "NIELLEZ", "Prenom" : "Christophe" }
{ "IdSportif" : 29, "Nom" : "ROUSSEL", "Prenom" : "Nadège" }
{ "IdSportif" : 30, "Nom" : "SCHINK", "Prenom" : "Nicolas" }
{ "IdSportif" : 31, "Nom" : "STEMPUT", "Prenom" : "Mathieu" }
{ "IdSportif" : 32, "Nom" : "VAN CAUTER", "Prenom" : "Vincent" }
{ "IdSportif" : 33, "Nom" : "RAMPNOUX", "Prenom" : "Jean françois" }
{ "IdSportif" : 34, "Nom" : "LHERPINIERE", "Prenom" : "Olivier" }
{ "IdSportif" : 35, "Nom" : "TANQUE", "Prenom" : "Yann" }
{ "IdSportif" : 36, "Nom" : "DJELOUDANE", "Prenom" : "Zinedine" }
{ "IdSportif" : 37, "Nom" : "LAZZARI", "Prenom" : "Magali" }
{ "IdSportif" : 38, "Nom" : "VASSEMON", "Prenom" : "Laurent" }
{ "IdSportif" : 39, "Nom" : "MOREL", "Prenom" : "Mathieu" }
{ "IdSportif" : 40, "Nom" : "VALIN", "Prenom" : "Yann" }
{ "IdSportif" : 41, "Nom" : "DELOVINA", "Prenom" : "Elina" }
{ "IdSportif" : 42, "Nom" : "LEHOUX", "Prenom" : "Bruno" }
{ "IdSportif" : 43, "Nom" : "MATHIEU", "Prenom" : "Denise" }
{ "IdSportif" : 44, "Nom" : "ADIBO", "Prenom" : "Senamé" }
{ "IdSportif" : 45, "Nom" : "CHAVANT", "Prenom" : "Christophe" }
{ "IdSportif" : 46, "Nom" : "DAUXIAN", "Prenom" : "Cédric" }
{ "IdSportif" : 47, "Nom" : "GROEN", "Prenom" : "Céline" }
{ "IdSportif" : 48, "Nom" : "HEDDI", "Prenom" : "Zohra" }
{ "IdSportif" : 49, "Nom" : "JOUVE", "Prenom" : "Sandra" }
{ "IdSportif" : 50, "Nom" : "KALOMBO", "Prenom" : "Yannick" }
{ "IdSportif" : 51, "Nom" : "LOPEZ", "Prenom" : "Thibaud" }
{ "IdSportif" : 52, "Nom" : "DANDOIS", "Prenom" : "Régis" }
...

```

**b. Quels sont les gymnases de “Villetaneuse” ou de “Sarcelles” qui ont une surface de plus de 400 m2 ?**

```
db.getCollection('Gymnases').find(

{
 Ville: { $in: ["Villetaneuse", "Sarcelles"] },
 Surface: { $gt: 400 }
}
)

```

**Résultat :**

Fetches 0 record(s) in 7ms

**c. Quels sont les sportifs (identifiant et nom) qui pratiquent du hand ball ?**

```
db.getCollection('Sportifs').find(
{ "Sports.Jouer": "Hand ball" },
{ IdSportif: 1, Nom: 1, _id: 0 }
)

```

**Résultat (Extrait) :**

```
/* 1 */

{
 "IdSportif" : 6,
 "Nom" : "RETALDI"
}

```

```
}

/* 2 */
{
 "IdSportif" : 19,
 "Nom" : "TESTEMONT"
}

/* 3 */
{
 "IdSportif" : 20,
 "Nom" : "TIENER"
}

/* 4 */
{
 "IdSportif" : 25,
 "Nom" : "MARTORA"
}

/* 5 */
{
 "IdSportif" : 26,
 "Nom" : "MICHEL"
}

/* 6 */
{
 "IdSportif" : 27,
 "Nom" : "NIELLEZ"
}

/* 7 */
{
 "IdSportif" : 28,
 "Nom" : "RABAHI"
}
```

```

/* 8 */
{
 "IdSportif" : 29,
 "Nom" : "ROUSSEL"
}

/* 9 */
{
 "IdSportif" : 30,
 "Nom" : "SCHINK"
}

/* 10 */
{
 "IdSportif" : 31,
 "Nom" : "STEMPUT"
}

/* 11 */
{
 "IdSportif" : 32,
 "Nom" : "VAN CAUTER"
}

/* 12 */
{
 "IdSportif" : 33,
 "Nom" : "RAMPNOUX"
}

/* 13 */
{
 "IdSportif" : 34,
 "Nom" : "LHERPINIERE"
}

```

```
/* 14 */
{
 "IdSportif" : 35,
 "Nom" : "TANQUE"
}

/* 15 */
{
 "IdSportif" : 38,
 "Nom" : "VASSEMON"
}

/* 16 */
{
 "IdSportif" : 39,
 "Nom" : "MOREL"
}

/* 17 */
{
 "IdSportif" : 40,
 "Nom" : "VALIN"
}

/* 18 */
{
 "IdSportif" : 43,
 "Nom" : "MATHIEU"
}

/* 19 */
{
 "IdSportif" : 51,
 "Nom" : "LOPEZ"
}

/* 20 */
```

```

{
 "IdSportif" : 60,
 "Nom" : "TIZEGHAT"
}

/* 21 */
{
 "IdSportif" : 67,
 "Nom" : "VONTHRON"
}

/* 22 */
{
 "IdSportif" : 68,
 "Nom" : "REGNAULD"
}

/* 23 */
{
 "IdSportif" : 69,
 "Nom" : "MARIE"
}

/* 24 */
{
 "IdSportif" : 72,
 "Nom" : "MARTEL"
}

/* 25 */
{
 "IdSportif" : 75,
 "Nom" : "SARRAZIN"
}

/* 26 */
{

```

```

 "IdSportif" : 83,
 "Nom" : "LEROUX"
 }

/* 27 */
{
 "IdSportif" : 84,
 "Nom" : "SALLAI"
}

/* 28 */
{
 "IdSportif" : 89,
 "Nom" : "RIQUIER"
}

/* 29 */
{
 "IdSportif" : 94,
 "Nom" : "LEROUX"
}

/* 30 */
{
 "IdSportif" : 98,
 "Nom" : "RICHARD"
}

/* 31 */
{
 "IdSportif" : 100,
 "Nom" : "LESOIF"
}

/* 32 */
{
 "IdSportif" : 101,

```

```

 "Nom" : "SWERTVAEGER"
 }

/* 33 */
{
 "IdSportif" : 104,
 "Nom" : "SEHIER"
}

/* 34 */
{
 "IdSportif" : 105,
 "Nom" : "STILO"
}

/* 35 */
{
 "IdSportif" : 109,
 "Nom" : "SAUVAGE"
}

/* 36 */
{
 "IdSportif" : 110,
 "Nom" : "ROULLAND"
}

/* 37 */
{
 "IdSportif" : 111,
 "Nom" : "LESIEUR"
}

/* 38 */
{
 "IdSportif" : 112,
 "Nom" : "LUYCKX"
}

```

```

}

/* 39 */
{
 "IdSportif" : 114,
 "Nom" : "MICHEL"
}

/* 40 */
{
 "IdSportif" : 123,
 "Nom" : "LETHIMONNIER"
}

/* 41 */
{
 "IdSportif" : 124,
 "Nom" : "SWERTVAEGER"
}

/* 42 */
{
 "IdSportif" : 135,
 "Nom" : "MANCEL"
}

/* 43 */
{
 "IdSportif" : 138,
 "Nom" : "MADELAINE"
}

/* 44 */
{
 "IdSportif" : 151,
 "Nom" : "HENRY"
}

```



***d. Dans quels gymnases et quels jours y a-t-il des séances de hand ball ?***

```
db.getCollection('Gymnases').find(
 { "Seances.Libelle": "Hand ball" },
 { NomGymnase: 1, "Seances.Jour": 1, _id: 0 }
```

**) Résultat :**

```
/* 1 */
```

```
{
 "NomGymnase" : "PAUL ELUARD",
 "Seances" : [
 {
 "Jour" : "Samedi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Lundi"
 },
 {
 "Jour" : "Dimanche"
 },
 {
 "Jour" : "Dimanche"
 },
 {
 "Jour" : "mardi"
```

```

 },
 {
 "Jour" : "mercredi"
 },
 {
 "Jour" : "Samedi"
 },
 {
 "Jour" : "Samedi"
 },
 {
 "Jour" : "Samedi"
 },
 {
 "Jour" : "jeudi"
 },
 {
 "Jour" : "lundi"
 },
 {
 "Jour" : "lundi"
 },
 {
 "Jour" : "lundi"
 },
 {
 "Jour" : "lundi"
 },
 {
 "Jour" : "mercredi"
 }
]

}

/* 2 */

{
 "NomGymnase" : "GARCIA LORCA",

```

```

 "Seances" : [
 {
 "Jour" : "samedi"
 },
 {
 "Jour" : "vendredi"
 },
 {
 "Jour" : "dimanche"
 },
 {
 "Jour" : "dimanche"
 },
 {
 "Jour" : "mardi"
 },
 {
 "Jour" : "mardi"
 }
]
 }
}

```

***e. Dans quels gymnases peut-on jouer au hockey le mercredi après 15H ?***

```

db.getCollection('Gymnases').find(
 {
 Seances: {
 $elemMatch: {
 Libelle: "Hockey",
 Jour: "mercredi",
 Horaire: { $gt: 15 }
 }
 }
 },
 { NomGymnase: 1, _id: 0 }
)

```

**Résultat:**

```

/* 1 */

{
 "NomGymnase" : "PAUL ELUARD"
}

/* 2 */

```

```

{
 "NomGymnase" : "PAUL ELUARD"
}

/* 3 */

{
 "NomGymnase" : "RAVEL"
}

/* 4 */

{
 "NomGymnase" : "SAMOURAI"
}

/* 5 */

{
 "NomGymnase" : "PABLO NERUDA"
}

/* 6 */

{
 "NomGymnase" : "LUMIERES"
}

/* 7 */

{
 "NomGymnase" : "SIMON"
}

/* 8 */

{
 "NomGymnase" : "PASCAL"
}

```

***f. Quels sportifs (identifiant et nom) ne pratiquent aucun sport ?***

```

db.getCollection('Sportifs').find(
{
 $and: [
 { "Sports.Jouer": { $exists: false } },

```

```

 { "Sports.Entraîner": { $exists: false } },
 { "Sports.Arbitrer": { $exists: false } }
]
},
{ IdSportif: 1, Nom: 1, _id: 0 }
)

```

Résultat :

```
````javascript
```

```

{ "IdSportif" : 81, "Nom" : "DROULLON" }
{ "IdSportif" : 116, "Nom" : "BELUAU" }
{ "IdSportif" : 117, "Nom" : "FERREIRA" }
{ "IdSportif" : 131, "Nom" : "GALLOIS" }
{ "IdSportif" : 145, "Nom" : "LENEVEU" }
{ "IdSportif" : 146, "Nom" : "DABON" }
{ "IdSportif" : 147, "Nom" : "CLERICE" }
{ "IdSportif" : 148, "Nom" : "COMES" }
{ "IdSportif" : 150, "Nom" : "BELZ" }
````

```

### ***g. Quels gymnases n'ont pas de séances le dimanche ?***

```

db.getCollection('Gymnases').find(
 { "Seances.Jour": { $ne: "dimanche" } },
 { NomGymnase: 1, _id: 0 }
)

```

Résultat:

```
/* 1 */
```

```

{
 "NomGymnase" : "PAUL ELUARD"
}

```

```
/* 2 */
```

```

{
 "NomGymnase" : "SAINT EXUPERY"
}

```

```
/* 3 */

{

 "NomGymnase" : "PAUL ELUARD"

}
```

```
/* 4 */

{

 "NomGymnase" : "BRASSENS"

}
```

```
/* 5 */

{

 "NomGymnase" : "VERLAINE"

}
```

```
/* 6 */

{

 "NomGymnase" : "JULES FERRY"

}
```

```
/* 7 */

{

 "NomGymnase" : "CAMUS"

}
```

```
/* 8 */

{

 "NomGymnase" : "LAMARTINE"

}
```

```
/* 9 */

{

 "NomGymnase" : "BREL"

}
```

```
/* 10 */

{
```

```

 "NomGymnase" : "SAMOURAI"
 }

/* 11 */
{
 "NomGymnase" : "PABLO NERUDA"
}

/* 12 */
{
 "NomGymnase" : "LUMIERES"
}

/* 13 */
{
 "NomGymnase" : "SIMON"
}

/* 14 */
{
 "NomGymnase" : "BARBARA"
}

/* 15 */
{
 "NomGymnase" : "ARAGON"
}

/* 16 */
{
 "NomGymnase" : "BELFEGOR"
}

/* 17 */
{
 "NomGymnase" : "DOLTO"
}

```

```
/* 18 */
```

```
{
 "NomGymnase" : "MERMOZ"
}
```

```
/* 19 */
```

```
{
 "NomGymnase" : "PASCAL"
}
```

### ***h. Quels gymnases ne proposent que des séances de basket ball ou de volley ball ?***

```
db.getCollection('Gymnases').find({ "Seances.Libelle": { $not: { $elemMatch: { $nin:
["Basket ball", "Volley ball"] } } } })
```

#### **Résultat:**

```
/* 1 */
```

```
{
 "_id" : "566eec69662b388eba464299",
 "IdGymnase" : 1,
 "NomGymnase" : "PAUL ELUARD",
 "Adresse" : "2 rue des pépines",
 "Ville" : "STAINS",
 "Surface" : 200,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Samedi",
 "Horaire" : 9.0,
 "Duree" : 60,
 "Libelle" : "Basket ball"
 },
 {
 "IdSportifEntraîneur" : 1,
 "Jour" : "Lundi",
 "Horaire" : 9.0,
 "Duree" : 60,
 "Libelle" : "Hand ball"
 },
],
}
```



```

{
 "IdSportifEntraîneur" : 1,
 "Jour" : "Lundi",
 "Horaire" : 10.0,
 "Duree" : 60,
 "Libelle" : "Hand ball"
},
{
 "IdSportifEntraîneur" : 1,
 "Jour" : "Lundi",
 "Horaire" : 11.3,
 "Duree" : 60,
 "Libelle" : "Hand ball"
},
{
 "IdSportifEntraîneur" : 1,
 "Jour" : "Lundi",
 "Horaire" : 14.0,
 "Duree" : 90,
 "Libelle" : "Hand ball"
},
{
 "IdSportifEntraîneur" : 1,
 "Jour" : "lundi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
},
{
 "IdSportifEntraîneur" : 1,
 "Jour" : "Lundi",
 "Horaire" : 19.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
},
{
 "IdSportifEntraîneur" : 2,

```

```

 "Jour" : "Dimanche",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "Dimanche",
 "Horaire" : 19.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "mardi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "mercredi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "Samedi",
 "Horaire" : 15.3,
 "Duree" : 60,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "Samedi",
 "Horaire" : 16.3,

```

```

 "Duree" : 60,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 2,
 "Jour" : "Samedi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "jeudi",
 "Horaire" : 20.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "lundi",
 "Horaire" : 14.0,
 "Duree" : 60,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "lundi",
 "Horaire" : 18.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "lundi",
 "Horaire" : 19.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 }

```

```

 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "lundi",
 "Horaire" : 20.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 2 */
{
 "_id" : "566eec69662b388eba46429a",
 "IdGymnase" : 2,
 "NomGymnase" : "ARAGON",
 "Adresse" : "Place du Chartres",
 "Ville" : "MONTMORENCY",
 "Surface" : 450,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]
}
}

```

```

/* 3 */
{
 "_id" : "566eec69662b388eba46429b",
 "IdGymnase" : 3,
 "NomGymnase" : "SAINT EXUPERY",
 "Adresse" : "47 bvd des brumes",
 "Ville" : "PIERREFITTE",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Mercredi",
 "Horaire" : 11.0,
 "Duree" : 30,
 "Libelle" : "Basket ball"
 },
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "jeudi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]
}

/* 4 */
{
 "_id" : "566eec69662b388eba46429c",
 "IdGymnase" : 4,

```

```

 "NomGymnase" : "PAUL ELUARD",
 "Adresse" : "Allée J.B. Lulli",
 "Ville" : "SARCELLES",
 "Surface" : 500,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Vendredi",
 "Horaire" : 10.0,
 "Duree" : 30,
 "Libelle" : "Basket ball"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "mercredi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 }
]
}

```

```

/* 5 */

```

```

{
 "_id" : "566eec69662b388eba46429d",
 "IdGymnase" : 5,
 "NomGymnase" : "BRASSENS",
 "Adresse" : "153 square Loliot",
 "Ville" : "SARCELLES",
 "Surface" : 620,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 }
]
}

```

```

 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "jeudi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 }
]
}

/* 6 */
{
 "_id" : "566eec69662b388eba46429e",
 "IdGymnase" : 6,
 "NomGymnase" : "VERLAINE",
 "Adresse" : "14 bvd Serrault",
 "Ville" : "STAINS",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "vendredi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "jeudi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

```

```

/* 7 */
{
 "_id" : "566eec69662b388eba46429f",
 "IdGymnase" : 7,
 "NomGymnase" : "JULES FERRY",
 "Adresse" : "45 rue de la gare",
 "Ville" : "PIERREFITTE",
 "Surface" : 360
}

```

```

/* 8 */
{
 "_id" : "566eec69662b388eba4642a0",
 "IdGymnase" : 8,
 "NomGymnase" : "PREVERT",
 "Adresse" : "12 rue des collines",
 "Ville" : "MONTMORENCY",
 "Surface" : 420,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "vendredi",
 "Horaire" : 19.0,

```



```

 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "samedi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "vendredi",
 "Horaire" : 14.0,
 "Duree" : 120,
 "Libelle" : "Hockey"
 }
]
}

/* 9 */
{
 "_id" : "566eec69662b388eba4642a1",
 "IdGymnase" : 9,
 "NomGymnase" : "CAMUS",
 "Adresse" : "3 esplanade des quatrans",
 "Ville" : "SARCELLES",
 "Surface" : 620,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "samedi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 }
]
}

```

```

}

/* 10 */

{
 "_id" : "566eec69662b388eba4642a2",
 "IdGymnase" : 10,
 "NomGymnase" : "RIMBAUD",
 "Adresse" : "140 bvd Diderot",
 "Ville" : "STAINS",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "samedi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "dimanche",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 11 */

{

```

```

 "_id" : "566eec69662b388eba4642a3",
 "IdGymnase" : 11,
 "NomGymnase" : "LAMARTINE",
 "Adresse" : "7 rue de la souris verte",
 "Ville" : "PIERREFITTE",
 "Surface" : 300
}

```

```

/* 12 */

```

```

{
 "_id" : "566eec69662b388eba4642a4",
 "IdGymnase" : 12,
 "NomGymnase" : "MOZART",
 "Adresse" : "6 Allée Rosana",
 "Ville" : "MONTMORENCY",
 "Surface" : 480,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]
}

```

```

/* 13 */

```

```

{
 "_id" : "566eec69662b388eba4642a5",
 "IdGymnase" : 13,
 "NomGymnase" : "RAVEL",
 "Adresse" : "Place aux pommes",
 "Ville" : "STAINS",
 "Surface" : 200,
 "Seances" : [
 {

```

```

 "IdSportifEntraîneur" : 60,
 "Jour" : "dimanche",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "mercredi",
 "Horaire" : 20.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "lundi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 14 */
{
 "_id" : "566eec69662b388eba4642a6",
 "IdGymnase" : 14,
 "NomGymnase" : "CHOPIN",
 "Adresse" : "23 rue Carafelli",
 "Ville" : "MONTMORENCY",
 "Surface" : 500,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Mardi",
 "Horaire" : 10.0,
 "Duree" : 60,

```

```

 "Libelle" : "Basket ball"
 },
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]
}

```

```

/* 15 */

```

```

{
 "_id" : "566eec69662b388eba4642a7",
 "IdGymnase" : 15,
 "NomGymnase" : "BREL",
 "Adresse" : "4 rue de la miséricorde",
 "Ville" : "PIERREFITTE",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 }
]
}

```

```

/* 16 */

```

```

{
 "_id" : "566eec69662b388eba4642a8",
 "IdGymnase" : 16,
 "NomGymnase" : "SAMOURAI",
 "Adresse" : "4 Allée des pendules",

```

```

"Ville" : "SARCELLES",
"Surface" : 600,
"Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "Lundi",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "Lundi",
 "Horaire" : 18.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "lundi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "Lundi",
 "Horaire" : 20.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]

```

```

 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "mercredi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 }
]
}

/* 17 */
{
 "_id" : "566eec69662b388eba4642a9",
 "IdGymnase" : 17,
 "NomGymnase" : "GARCIA LORCA",
 "Adresse" : "45 bvd des Comes",
 "Ville" : "STAINS",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "samedi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "vendredi",
 "Horaire" : 17.3,
 "Duree" : 120,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",

```

```

 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "dimanche",
 "Horaire" : 18.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 3,
 "Jour" : "mardi",
 "Horaire" : 20.0,
 "Duree" : 30,
 "Libelle" : "Hand ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mardi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 18 */
{
 "_id" : "566eec69662b388eba4642aa",
 "IdGymnase" : 18,
 "NomGymnase" : "PABLO NERUDA",
 "Adresse" : "6 rue saint Jean",
 "Ville" : "PIERREFITTE",
 "Surface" : 450,
 "Seances" : [

```



```

 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "mardi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 14.0,
 "Duree" : 120,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 16.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 19 */
{
 "_id" : "566eec69662b388eba4642ab",
 "IdGymnase" : 19,
 "NomGymnase" : "COCTEAU",
 "Adresse" : "45 bis rue du moulin rouge",

```

```

"Ville" : "MONTMORENCY",
"Surface" : 500,
"Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "dimanche",
 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 }
]
}

```

```

/* 20 */
{
 "_id" : "566eec69662b388eba4642ac",
 "IdGymnase" : 20,
 "NomGymnase" : "LUMIERES",
 "Adresse" : "78 rue Vendôme",
 "Ville" : "SARCELLES",
 "Surface" : 400,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "mercredi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 }
]
}

```

```

/* 21 */
{
 "_id" : "566eec69662b388eba4642ad",
 "IdGymnase" : 21,
 "NomGymnase" : "SIMON",

```

```

"Adresse" : "8 bvd général de Gaulle",
"Ville" : "STAINS",
"Surface" : 400,
"Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "lundi",
 "Horaire" : 16.3,
 "Duree" : 30,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 60,
 "Jour" : "mardi",
 "Horaire" : 19.0,
 "Duree" : 60,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 17.0,
 "Duree" : 30,
 "Libelle" : "Hockey"
 }
]
}

```

```

/* 22 */
{
 "_id" : "566eec69662b388eba4642ae",
 "IdGymnase" : 22,
 "NomGymnase" : "BARBARA",
 "Adresse" : "45 rue du bossu",
 "Ville" : "SAINT DENIS",
 "Surface" : 500,
 "Seances" : [

```

```

 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "mardi",
 "Horaire" : 10.0,
 "Duree" : 30,
 "Libelle" : "Volley ball"
 }
]
}

```

/\* 23 \*/

```

{
 "_id" : "566eec69662b388eba4642af",
 "IdGymnase" : 23,
 "NomGymnase" : "ARAGON",
 "Adresse" : "10 Bvd Lenoir",
 "Ville" : "SAINT DENIS",
 "Surface" : 520
}

```

/\* 24 \*/

```

{
 "_id" : "566eec69662b388eba4642b0",
 "IdGymnase" : 24,
 "NomGymnase" : "BELFEGOR",
 "Adresse" : "Place de Gaulle",
 "Ville" : "SAINT DENIS",
 "Surface" : 450,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Jeudi",
 "Horaire" : 9.0,
 "Duree" : 90,
 "Libelle" : "Basket ball"
 },
 {

```

```

 "IdSportifEntraîneur" : 57,

 "Jour" : "mercredi",

 "Horaire" : 10.0,

 "Duree" : 90,

 "Libelle" : "Volley ball"

 }

]

}

/* 25 */

{

 "_id" : "566eec69662b388eba4642b1",

 "IdGymnase" : 25,

 "NomGymnase" : "DOLTO",

 "Adresse" : "3 square Plaisir",

 "Ville" : "VILLETANEUSE",

 "Surface" : 620,

 "Seances" : [

 {

 "IdSportifEntraîneur" : 149,

 "Jour" : "Dimanche",

 "Horaire" : 18.0,

 "Duree" : 60,

 "Libelle" : "Basket ball"

 }

]

}

/* 26 */

{

 "_id" : "566eec69662b388eba4642b2",

 "IdGymnase" : 26,

 "NomGymnase" : "MERMOZ",

 "Adresse" : "41 rue des ponts",

 "Ville" : "VILLETANEUSE",

 "Surface" : 600

}

```

```

/* 27 */
{
 "_id" : "566eec69662b388eba4642b3",
 "IdGymnase" : 27,
 "NomGymnase" : "PASCAL",
 "Adresse" : "20 rue de la pirogue",
 "Ville" : "VILLETANEUSE",
 "Surface" : 350,
 "Seances" : [
 {
 "IdSportifEntraîneur" : 57,
 "Jour" : "jeudi",
 "Horaire" : 10.0,
 "Duree" : 90,
 "Libelle" : "Volley ball"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 14.0,
 "Duree" : 120,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mercredi",
 "Horaire" : 17.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

/* 28 */
{
 "_id" : "566eec69662b388eba4642b4",

```

```

"IdGymnase" : 28,
"NomGymnase" : "BLAISE PASCAL",
"Adresse" : "2bis rue de la moulerie",
"Ville" : "GARGES",
"Surface" : 400,
"Seances" : [
 {
 "IdSportifEntraîneur" : 149,
 "Jour" : "Lundi",
 "Horaire" : 9.0,
 "Duree" : 30,
 "Libelle" : "Basket ball"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "dimanche",
 "Horaire" : 14.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "dimanche",
 "Horaire" : 15.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "dimanche",
 "Horaire" : 16.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 6,
 "Jour" : "dimanche",

```

```

 "Horaire" : 17.0,
 "Duree" : 60,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "mardi",
 "Horaire" : 18.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "samedi",
 "Horaire" : 18.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 },
 {
 "IdSportifEntraîneur" : 7,
 "Jour" : "vendredi",
 "Horaire" : 18.0,
 "Duree" : 90,
 "Libelle" : "Hockey"
 }
]
}

```

#### ***i. Quels sont les entraîneurs qui sont aussi joueurs ?***

```

db.getCollection('Sportifs').find(
{
 "Sports.Entraîneur": { $exists: true },
 "Sports.Jouer": { $exists: true }
},
{ IdSportif: 1, Nom: 1, Prenom: 1, _id: 0 }
)

```

#### **Résultat (Extrait) :**

```

/* 1 */
{
 "IdSportif" : 1,

```



```

 "Nom" : "BOUTAHAR",
 "Prenom" : "Abderahim"
 }

/* 2 */
{
 "IdSportif" : 2,
 "Nom" : "KERVADEC",
 "Prenom" : "Yann"
}

/* 3 */
{
 "IdSportif" : 3,
 "Nom" : "HUE",
 "Prenom" : "Pascale"
}

/* 4 */
{
 "IdSportif" : 4,
 "Nom" : "DORLEANS",
 "Prenom" : "Jean-michel"
}

/* 5 */
{
 "IdSportif" : 6,
 "Nom" : "RETALDI",
 "Prenom" : "Sophie"
}

/* 6 */
{
 "IdSportif" : 7,
 "Nom" : "GOMEZ",
 "Prenom" : "Diego"
}

```

```

}

/* 7 */
{
 "IdSportif" : 29,
 "Nom" : "ROUSSEL",
 "Prenom" : "Nadège"
}

/* 8 */
{
 "IdSportif" : 30,
 "Nom" : "SCHINK",
 "Prenom" : "Nicolas"
}

/* 9 */
{
 "IdSportif" : 31,
 "Nom" : "STEMPUT",
 "Prenom" : "Mathieu"
}

/* 10 */
{
 "IdSportif" : 32,
 "Nom" : "VAN CAUTER",
 "Prenom" : "Vincent"
}

/* 11 */
{
 "IdSportif" : 35,
 "Nom" : "TANQUE",
 "Prenom" : "Yann"
}

```

```
/* 12 */

{
 "IdSportif" : 36,
 "Nom" : "DJELOUDANE",
 "Prenom" : "Zinedine"
}
```

```
/* 13 */

{
 "IdSportif" : 38,
 "Nom" : "VASSEMON",
 "Prenom" : "Laurent"
}
```

```
/* 14 */

{
 "IdSportif" : 40,
 "Nom" : "VALIN",
 "Prenom" : "Yann"
}
```

```
/* 15 */

{
 "IdSportif" : 48,
 "Nom" : "HEDDI",
 "Prenom" : "Zohra"
}
```

```
/* 16 */

{
 "IdSportif" : 50,
 "Nom" : "KALOMBO",
 "Prenom" : "Yannick"
}
```

```
/* 17 */

{
```

```

 "IdSportif" : 56,
 "Nom" : "GUERRAOUI",
 "Prenom" : "Zohra"
 }

/* 18 */
{
 "IdSportif" : 57,
 "Nom" : "BOISSEAU",
 "Prenom" : "Eric"
}

/* 19 */
{
 "IdSportif" : 58,
 "Nom" : "GUIGUI",
 "Prenom" : "Vincent"
}

/* 20 */
{
 "IdSportif" : 59,
 "Nom" : "CLERICE",
 "Prenom" : "Stéphanie"
}

/* 21 */
{
 "IdSportif" : 60,
 "Nom" : "TIZEGHAT",
 "Prenom" : "Benamar"
}

/* 22 */
{
 "IdSportif" : 61,
 "Nom" : "LAZARRE",

```

```

 "Prenom" : "Jean"
 }

/* 23 */

{
 "IdSportif" : 149,
 "Nom" : "BELZ",
 "Prenom" : "Sylvianne"
}

/* 24 */

{
 "IdSportif" : 151,
 "Nom" : "HENRY",
 "Prenom" : "Maël"
}

```

#### ***j. Quels sont les sportifs qui sont des conseillers ?***

```

db.Sportifs.aggregate([
 { $lookup: { from: "Sportifs", localField: "IdSportif",
 foreignField: "IdSportifConseiller", as: "MesConseilles" } },
 { $match: { "MesConseilles.0": { $exists: true } } },
 { $project: { Nom: 1, Prenom: 1, _id: 0 } }
])

```

Résultat :

```
````javascript
```

```

{ "Nom": "BOUTAHAR", "Prenom": "Abderahim" }
{ "Nom": "KERVADEC", "Prenom": "Yann" }
{ "Nom": "HUE", "Prenom": "Pascale" }
{ "Nom": "DORLEANS", "Prenom": "Jean-michel" }
{ "Nom": "COMES", "Prenom": "Sylvie" }
{ "Nom": "RETALDI", "Prenom": "Sophie" }
{ "Nom": "GOMEZ", "Prenom": "Diego" }
{ "Nom": "ANTUNES", "Prenom": "Gaëlle" }
{ "Nom": "CAILLIOT", "Prenom": "Stéphanie" }
{ "Nom": "LE BOUCHER", "Prenom": "Denis" }
{ "Nom": "ADIBO", "Prenom": "Senamé" }
{ "Nom": "RICHARD", "Prenom": "William" }
````

```

***k. Pour le sportif "Kervadec" quel est le nom de son conseiller ?***

```
var kervadec = db.Sportifs.findOne({Nom: "KERVADEC"});
db.Sportifs.find({IdSportif: kervadec.IdSportifConseiller}, {Nom: 1, _id: 0})
```

**Résultat:**

```
/* 1 */

{

 "Nom" : "BOUTAHAR"

}
```

***l. Quels entraîneurs entraînent du hand ball et du basket ball ?***

```
db.Sportifs.find({"Sports.Entraîner": {$all: ["Hand ball", "Basket ball"]}}, {Nom:1,
_id:0})
```

**Résultat:**

```
/* 1 */

{

 "Nom" : "BOUTAHAR"

}
```

```
/* 2 */

{

 "Nom" : "KERVADEC"

}
```

```
/* 3 */

{

 "Nom" : "HUE"

}
```

```
/* 4 */

{

 "Nom" : "HENRY"

}
```

```
}
```

***m. Quels sont les couples de sportifs (identifiant et nom et prénom de chaque) de même âge ?***

```
db.getCollection('Sportifs').aggregate([
 {
 $lookup: {
 from: "Sportifs",
 localField: "Age",
 foreignField: "Age",
 as: "MemeAge"
 }
 },
 { $unwind: "$MemeAge" },
 { $match: { $expr: { $lt: ["$IdSportif", "$MemeAge.IdSportif"] } } },
 {
 $project: {
 _id: 0,
 Id1: "$IdSportif", Nom1: "$Nom", Prenom1: "$Prenom",
 Id2: "$MemeAge.IdSportif", Nom2: "$MemeAge.Nom", Prenom2: "$MemeAge.Prenom",
 Age: 1
 }
 }
])
```

**Résultat :**

```
/* 1 */
```

```
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 6,
 "Nom2" : "RETALDI",
 "Prenom2" : "Sophie"
}
```

```
/* 2 */
```

```
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 63,
 "Nom2" : "AMAND",
 "Prenom2" : "Patrick"
}
```

```

/* 3 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 64,
 "Nom2" : "LANOE",
 "Prenom2" : "Françoise"
}

```

```

/* 4 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 65,
 "Nom2" : "CHESNIER",
 "Prenom2" : "Marc"
}

```

```

/* 5 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 66,
 "Nom2" : "DURIEU",
 "Prenom2" : "Loïc"
}

```

```

/* 6 */
{
 "Age" : 30,

```



```

 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 68,
 "Nom2" : "REGNAULD",
 "Prenom2" : "Jeanne"
 }

```

```

/* 7 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 70,
 "Nom2" : "BELLAMY",
 "Prenom2" : "Norbert"
}

```

```

/* 8 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 71,
 "Nom2" : "DELAROCHE",
 "Prenom2" : "Bertrand"
}

```

```

/* 9 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 73,

```

```

 "Nom2" : "DALLIER",
 "Prenom2" : "Didier"
 }

/* 10 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 74,
 "Nom2" : "AUVRAY",
 "Prenom2" : "Alain"
}

/* 11 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 76,
 "Nom2" : "HALGATTE",
 "Prenom2" : "Claude"
}

/* 12 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 77,
 "Nom2" : "HEON",
 "Prenom2" : "Philippe"
}

```

```

/* 13 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 78,
 "Nom2" : "CHAUVIN",
 "Prenom2" : "Julien"
}

```

```

/* 14 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 79,
 "Nom2" : "HENRY",
 "Prenom2" : "Jacky"
}

```

```

/* 15 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 80,
 "Nom2" : "HOCHET",
 "Prenom2" : "Pierre"
}

```

```

/* 16 */
{
 "Age" : 30,
 "Id1" : 1,

```

```

 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 81,
 "Nom2" : "DROULLON",
 "Prenom2" : "Joëlle"
 }

```

```

/* 17 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 83,
 "Nom2" : "LEROUX",
 "Prenom2" : "André"
}

```

```

/* 18 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 85,
 "Nom2" : "HAMARD",
 "Prenom2" : "Romain"
}

```

```

/* 19 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 86,
 "Nom2" : "GALLOT",

```

```

 "Prenom2" : "Bernard"
 }

/* 20 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 87,
 "Nom2" : "COUESBOT",
 "Prenom2" : "Daniel"
}

/* 21 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 88,
 "Nom2" : "LEJEUNE",
 "Prenom2" : "Richard"
}

/* 22 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 89,
 "Nom2" : "RIQUIER",
 "Prenom2" : "Jean-pierre"
}

/* 23 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 90,
 "Nom2" : "DUREL",
 "Prenom2" : "Eric"
}

```

/\* 24 \*/

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 91,
 "Nom2" : "LECHEVALIER",
 "Prenom2" : "Patrick"
}

```

/\* 25 \*/

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 93,
 "Nom2" : "CAUCHARD",
 "Prenom2" : "Georges"
}

```

/\* 26 \*/

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",

```

```

 "Prenom1" : "Abderahim",
 "Id2" : 95,
 "Nom2" : "EUSTACHE",
 "Prenom2" : "Marcel"
 }

```

```

/* 27 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 96,
 "Nom2" : "JANY",
 "Prenom2" : "Claude"
}

```

```

/* 28 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 97,
 "Nom2" : "BONHOMME",
 "Prenom2" : "Bruno"
}

```

```

/* 29 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 98,
 "Nom2" : "RICHARD",
 "Prenom2" : "William"
}

```

```

}

/* 30 */

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 100,
 "Nom2" : "LESOIF",
 "Prenom2" : "Jacques"
}

```

```

/* 31 */

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 101,
 "Nom2" : "SWERTVAEGER",
 "Prenom2" : "Eric"
}

```

```

/* 32 */

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 102,
 "Nom2" : "DUVAL",
 "Prenom2" : "Alain"
}

```

```

/* 33 */

{

```



```

 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 103,
 "Nom2" : "LEMENOREL",
 "Prenom2" : "Claude"
}

```

```
/* 34 */
```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 104,
 "Nom2" : "SEHIER",
 "Prenom2" : "Dominique"
}

```

```
/* 35 */
```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 105,
 "Nom2" : "STILO",
 "Prenom2" : "Philippe"
}

```

```
/* 36 */
```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",

```

```

 "Id2" : 106,

 "Nom2" : "LE BANNARD",

 "Prenom2" : "G  rard"
 }

/* 37 */

{
 "Age" : 30,

 "Id1" : 1,

 "Nom1" : "BOUTAHAR",

 "Prenom1" : "Abderahim",

 "Id2" : 107,

 "Nom2" : "BORNE",

 "Prenom2" : "Jean-Yves"
}

/* 38 */

{
 "Age" : 30,

 "Id1" : 1,

 "Nom1" : "BOUTAHAR",

 "Prenom1" : "Abderahim",

 "Id2" : 108,

 "Nom2" : "LAURENCE",

 "Prenom2" : "Serge"
}

/* 39 */

{
 "Age" : 30,

 "Id1" : 1,

 "Nom1" : "BOUTAHAR",

 "Prenom1" : "Abderahim",

 "Id2" : 109,

 "Nom2" : "SAUVAGE",

 "Prenom2" : "Patrick"
}

```

```

/* 40 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 110,
 "Nom2" : "ROULLAND",
 "Prenom2" : "Christian"
}

```

```

/* 41 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 111,
 "Nom2" : "LESIEUR",
 "Prenom2" : "Michel"
}

```

```

/* 42 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 112,
 "Nom2" : "LUYCKX",
 "Prenom2" : "G  rard"
}

```

```

/* 43 */
{
 "Age" : 30,

```

```

 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 113,
 "Nom2" : "AVICE",
 "Prenom2" : "Benoît"
 }

```

```

/* 44 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 115,
 "Nom2" : "LEMOUSSU",
 "Prenom2" : "Laurent"
}

```

```

/* 45 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 116,
 "Nom2" : "BELUAU",
 "Prenom2" : "Gilberte"
}

```

```

/* 46 */

```

```

{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 117,

```

```

 "Nom2" : "FERREIRA",
 "Prenom2" : "Martine"
 }

/* 47 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 118,
 "Nom2" : "FROMENT",
 "Prenom2" : "Philippe"
}

/* 48 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 119,
 "Nom2" : "GUITON",
 "Prenom2" : "Jean-paul"
}

/* 49 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 120,
 "Nom2" : "LECOMTE",
 "Prenom2" : "Christian"
}

```

```

/* 50 */
{
 "Age" : 30,
 "Id1" : 1,
 "Nom1" : "BOUTAHAR",
 "Prenom1" : "Abderahim",
 "Id2" : 121,
 "Nom2" : "HUET",
 "Prenom2" : "Loïc"
}

```

#### ***n. Quelle est la moyenne d'âge des sportives qui pratiquent du basket ball ?***

```

db.Sportifs.aggregate([
 { $match: { Sexe: { $regex: /^f$/i }, "Sports.Jouer": "Basket ball" } },
 { $group: { _id: null, moyenneAge: { $avg: "$Age" } } }
])

```

*Rôle : Filtre les sportives (Sexe='F' ou 'f') jouant au basket, puis calcule la moyenne (\$avg) de leur âge.*

Résultat :

```
````javascript
```

```
{ "_id" : null, "MoyenneAge" : 24.6 }
```

```
````
```

#### ***o. Quels sont les sportifs les plus jeunes ?***

```

db.getCollection('Sportifs').aggregate([
 { $group: { _id: null, MinAge: { $min: "$Age" } } },
 {
 $lookup: {
 from: "Sportifs",
 localField: "MinAge",
 foreignField: "Age",
 as: "PlusJeunes"
 }
 },
 { $unwind: "$PlusJeunes" },
 { $project: { _id: 0, Nom: "$PlusJeunes.Nom", Prenom: "$PlusJeunes.Prenom", Age:
"$MinAge" } }
])

```

Résultat :

```
````javascript
```

```

{ "Nom" : "COMES", "Prenom" : "Sylvie", "Age" : 22 }
{ "Nom" : "BOYON", "Prenom" : "Julien", "Age" : 22 }
{ "Nom" : "COLOMB", "Prenom" : "Michaël", "Age" : 22 }
{ "Nom" : "LE BOUCHER", "Prenom" : "Denis", "Age" : 22 }
{ "Nom" : "ROUSSEL", "Prenom" : "Nadège", "Age" : 22 }
{ "Nom" : "STEMPUT", "Prenom" : "Mathieu", "Age" : 22 }
{ "Nom" : "DELOVINA", "Prenom" : "Elina", "Age" : 22 }
{ "Nom" : "KALOMBO", "Prenom" : "Yannick", "Age" : 22 }

```

```
{ "Nom" : "DANDOIS", "Prenom" : "Régis", "Age" : 22 }
{ "Nom" : "DEMMER", "Prenom" : "Thomas", "Age" : 22 }
{ "Nom" : "MARTEL", "Prenom" : "Bernard", "Age" : 22 }
{ "Nom" : "HERVIEU", "Prenom" : "Jean-François", "Age" : 22 }
{ "Nom" : "MICHEL", "Prenom" : "Danielle", "Age" : 22 }
````
```

**p. Quels sont les gymnases de "Stains" ou de "Montmorency" qui ont la plus grande surface ?**

```
db.Gymnases.find({Ville: {$in: ["Stains", "Montmorency"]}},
 {NomGymnase: 1, Surface: 1, _id: 0}).sort({Surface: -1}).limit(1)
```

**Résultat:**

Fetches 0 record(s) in 2ms

**q. Quels entraîneurs n'entraînent que du hand ball ou du basket ball ?**

```
db.Sportifs.find({"Sports.Entraîneur": {$exists: true,
 $not: {$elemMatch: {$nin: ["Hand ball", "Basket ball"]}}}})
```

**Résultat :**

```
/* 1 */
{
 "_id" : "566eec5f662b388eba46421e",
 "IdSportif" : 29,
 "Nom" : "ROUSSEL",
 "Prenom" : "Nadège",
 "Sexe" : "F",
 "Age" : 22,
 "IdSportifConseiller" : 5,
 "Sports" : {
 "Jouer" : [
 "Volley ball",
```

```

 "Hand ball",
 "Badmington",
 "Ping pong"
],
 "Arbitrer" : "Ping pong",
 "Entraîner" : "Ping pong"
}
}

```

```

/* 2 */

```

```

{
 "_id" : "566eec5f662b388eba46421f",
 "IdSportif" : 30,
 "Nom" : "SCHINK",
 "Prenom" : "Nicolas",
 "Sexe" : "M",
 "Age" : 24,
 "IdSportifConseiller" : 4,
 "Sports" : {
 "Jouer" : [
 "Volley ball",
 "Hand ball",
 "Ping pong",
 "Football"
],
 "Entraîner" : "Ping pong"
 }
}

```

```

/* 3 */

```

```

{
 "_id" : "566eec5f662b388eba464220",
 "IdSportif" : 31,
 "Nom" : "STEMPUT",
 "Prenom" : "Mathieu",
 "Sexe" : "M",
 "Age" : 22,

```



```

 "IdSportifConseiller" : 2,
 "Sports" : {
 "Jouer" : [
 "Volley ball",
 "Hand ball",
 "Badmington",
 "Football"
],
 "Entrainer" : "Ping pong"
 }
 }

/* 4 */
{
 "_id" : "566eec5f662b388eba464221",
 "IdSportif" : 32,
 "Nom" : "VAN CAUTER",
 "Prenom" : "Vincent",
 "Sexe" : "M",
 "Age" : 23,
 "IdSportifConseiller" : 3,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Volley ball",
 "Hand ball",
 "Badmington",
 "Ping pong",
 "Football"
],
 "Arbitrer" : "Ping pong",
 "Entrainer" : "Ping pong"
 }
}

/* 5 */
{

```

```

 "_id" : "566eec60662b388eba464225",
 "IdSportif" : 36,
 "Nom" : "DJELOUDANE",
 "Prenom" : "Zinedine",
 "Sexe" : "M",
 "Age" : 28,
 "IdSportifConseiller" : 1,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Volley ball",
 "Ping pong",
 "Football"
],
 "Entrainer" : "Badmington"
 }
 }
}

```

```

/* 6 */

```

```

{
 "_id" : "566eec60662b388eba464227",
 "IdSportif" : 38,
 "Nom" : "VASSEMON",
 "Prenom" : "Laurent",
 "Sexe" : "M",
 "Age" : 24,
 "IdSportifConseiller" : 1,
 "Sports" : {
 "Jouer" : [
 "Hand ball",
 "Badmington",
 "Football"
],
 "Entrainer" : "Ping pong"
 }
}

```

```

/* 7 */
{
 "_id" : "566eec60662b388eba464231",
 "IdSportif" : 48,
 "Nom" : "HEDDI",
 "Prenom" : "Zohra",
 "Sexe" : "F",
 "Age" : 23,
 "IdSportifConseiller" : 2,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Badmington",
 "Ping pong"
],
 "Entraîner" : "Badmington"
 }
}

```

```

/* 8 */
{
 "_id" : "566eec60662b388eba464233",
 "IdSportif" : 50,
 "Nom" : "KALOMBO",
 "Prenom" : "Yannick",
 "Sexe" : "M",
 "Age" : 22,
 "IdSportifConseiller" : 2,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Badmington",
 "Ping pong",
 "Football"
],
 "Entraîner" : "Badmington"
 }
}

```

```

}

/* 9 */
{
 "_id" : "566eec60662b388eba464239",
 "IdSportif" : 56,
 "Nom" : "GUERRAOUI",
 "Prenom" : "Zohra",
 "Sexe" : "F",
 "Age" : 25,
 "IdSportifConseiller" : 1,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Ping pong"
],
 "Entraîner" : "Badmington"
 }
}

/* 10 */
{
 "_id" : "566eec60662b388eba464296",
 "IdSportif" : 149,
 "Nom" : "BELZ",
 "Prenom" : "Sylvianne",
 "Sexe" : "F",
 "Age" : 27,
 "IdSportifConseiller" : 7,
 "Sports" : {
 "Jouer" : "Basket ball",
 "Arbitrer" : "Basket ball",
 "Entraîner" : "Basket ball"
 }
}

/* 11 */

```

```

{
 "_id" : "566eec60662b388eba464298",
 "IdSportif" : 151,
 "Nom" : "HENRY",
 "Prenom" : "Maël",
 "Sexe" : "M",
 "Age" : 25,
 "IdSportifConseiller" : 2,
 "Sports" : {
 "Jouer" : [
 "Basket ball",
 "Hand ball"
],
 "Arbitrer" : [
 "Basket ball",
 "Hand ball"
],
 "Entraîner" : [
 "Basket ball",
 "Hand ball"
]
 }
}

```

***r. Quels sont les couples de sportifs de même âge avec le même conseiller ?***

```

db.Sportifs.aggregate([
 { $group: { _id: { Age: "$Age", Conseiller: "$IdSportifConseiller" },
 sportifs: { $push: { IdSportif: "$IdSportif", Nom: "$Nom" } },
 count: { $sum: 1 } } },
 { $match: { count: { $gt: 1 }, "_id.Conseiller": {$ne: null} } }
])

```

Résultat :

```
````javascript
```

```

{ "_id" : { "Age" : 23, "Conseiller" : 2 },
  "sportifs" : [
    { "IdSportif" : 10, "Nom" : "BLANDET" },
    { "IdSportif" : 40, "Nom" : "VALIN" },
    { "IdSportif" : 48, "Nom" : "HEDDI" },
    { "IdSportif" : 54, "Nom" : "ELKABBADJ" },
    { "IdSportif" : 55, "Nom" : "FEROLDI" }
  ],
  "count" : 5.0
}

```

```

}
{ "_id" : { "Age" : 30, "Conseiller" : 8 },
  "sportifs" : [
    { "IdSportif" : 71, "Nom" : "DELAROCHE" },
    { "IdSportif" : 88, "Nom" : "LEJEUNE" },
    { "IdSportif" : 111, "Nom" : "LESIEUR" },
    { "IdSportif" : 116, "Nom" : "BELUAU" }
  ],
  "count" : 4.0
}
{ "_id" : { "Age" : 28, "Conseiller" : 1 },
  "sportifs" : [
    { "IdSportif" : 2, "Nom" : "KERVADEC" },
    { "IdSportif" : 36, "Nom" : "DJELOUDANE" }
  ],
  "count" : 2.0
}
{ "_id" : { "Age" : 25, "Conseiller" : 1 },
  "sportifs" : [
    { "IdSportif" : 3, "Nom" : "HUE" },
    { "IdSportif" : 19, "Nom" : "TESTEMONT" },
    { "IdSportif" : 56, "Nom" : "GUERRAOUI" },
    { "IdSportif" : 57, "Nom" : "BOISSEAU" }
  ],
  "count" : 4.0
}
{ "_id" : { "Age" : 26, "Conseiller" : 2 },
  "sportifs" : [
    { "IdSportif" : 67, "Nom" : "VONTHRON" },
    { "IdSportif" : 136, "Nom" : "LABOULAI" },
    { "IdSportif" : 137, "Nom" : "DUDOUIT" },
    { "IdSportif" : 144, "Nom" : "GRANDIDIER" }
  ],
  "count" : 4.0
}
{ "_id" : { "Age" : 22, "Conseiller" : 2 },
  "sportifs" : [
    { "IdSportif" : 21, "Nom" : "LE BOUCHER" },
    { "IdSportif" : 31, "Nom" : "STEMPUT" },
    { "IdSportif" : 50, "Nom" : "KALOMBO" }
  ],
  "count" : 3.0
}
{ "_id" : { "Age" : 23, "Conseiller" : 4 },
  "sportifs" : [
    { "IdSportif" : 9, "Nom" : "ANTUNES" },
    { "IdSportif" : 58, "Nom" : "GUIGUI" }
  ]
}

```

```

    ],
    "count" : 2.0
  }
  { "_id" : { "Age" : 23, "Conseiller" : 3 },
    "sportifs" : [
    { "IdSportif" : 26, "Nom" : "MICHEL" },
    { "IdSportif" : 32, "Nom" : "VAN CAUTER" }
    ],
    "count" : 2.0
  }
  { "_id" : { "Age" : 25, "Conseiller" : 3 },
    "sportifs" : [
    { "IdSportif" : 15, "Nom" : "CHAMPENOIS" },
    { "IdSportif" : 22, "Nom" : "LEDUFAUD" },
    { "IdSportif" : 146, "Nom" : "DABON" }
    ],
    "count" : 3.0
  }
  { "_id" : { "Age" : 26, "Conseiller" : 1 },
    "sportifs" : [
    { "IdSportif" : 138, "Nom" : "MADELAINES" },
    { "IdSportif" : 142, "Nom" : "CORNET" }
    ],
    "count" : 2.0
  }
  { "_id" : { "Age" : 24, "Conseiller" : 5 },
    "sportifs" : [
    { "IdSportif" : 49, "Nom" : "JOUVE" },
    { "IdSportif" : 51, "Nom" : "LOPEZ" }
    ],
    "count" : 2.0
  }
  { "_id" : { "Age" : 27, "Conseiller" : 7 },
    "sportifs" : [
    { "IdSportif" : 61, "Nom" : "LAZARRE" },
    { "IdSportif" : 149, "Nom" : "BELZ" }
    ],
    "count" : 2.0
  }
  { "_id" : { "Age" : 30, "Conseiller" : 1 },
    "sportifs" : [
    { "IdSportif" : 63, "Nom" : "AMAND" },
    { "IdSportif" : 77, "Nom" : "HEON" },
    { "IdSportif" : 80, "Nom" : "HOCHET" },
    { "IdSportif" : 81, "Nom" : "DROULLON" },
    { "IdSportif" : 104, "Nom" : "SEHIER" },
    { "IdSportif" : 106, "Nom" : "LE BANNARD" },

```

```

{ "IdSportif" : 115, "Nom" : "LEMOUSSU" },
{ "IdSportif" : 119, "Nom" : "GUITON" },
{ "IdSportif" : 128, "Nom" : "GIRONIE" },
{ "IdSportif" : 132, "Nom" : "DORON" }
],
"count" : 10.0
}
{ "_id" : { "Age" : 25, "Conseiller" : 2 },
  "sportifs" : [
{ "IdSportif" : 7, "Nom" : "GOMEZ" },
{ "IdSportif" : 20, "Nom" : "TIENER" },
{ "IdSportif" : 47, "Nom" : "GROEN" },
{ "IdSportif" : 145, "Nom" : "LENEVEU" },
{ "IdSportif" : 147, "Nom" : "CLERICE" },
{ "IdSportif" : 150, "Nom" : "BELZ" },
{ "IdSportif" : 151, "Nom" : "HENRY" }
],
"count" : 7.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 6 },
  "sportifs" : [
{ "IdSportif" : 73, "Nom" : "DALLIER" },
{ "IdSportif" : 120, "Nom" : "LECOMTE" },
{ "IdSportif" : 122, "Nom" : "BECQUET" },
{ "IdSportif" : 123, "Nom" : "LETHIMONNIER" }
],
"count" : 4.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 2 },
  "sportifs" : [
{ "IdSportif" : 64, "Nom" : "LANOE" },
{ "IdSportif" : 66, "Nom" : "DURIEU" },
{ "IdSportif" : 68, "Nom" : "REGNAULD" },
{ "IdSportif" : 79, "Nom" : "HENRY" },
{ "IdSportif" : 85, "Nom" : "HAMARD" },
{ "IdSportif" : 86, "Nom" : "GALLOT" },
{ "IdSportif" : 87, "Nom" : "COUESBOT" },
{ "IdSportif" : 90, "Nom" : "DUREL" },
{ "IdSportif" : 93, "Nom" : "CAUCHARD" },
{ "IdSportif" : 95, "Nom" : "EUSTACHE" },
{ "IdSportif" : 96, "Nom" : "JANY" },
{ "IdSportif" : 98, "Nom" : "RICHARD" },
{ "IdSportif" : 102, "Nom" : "DUVAL" },
{ "IdSportif" : 112, "Nom" : "LUYCKX" },
{ "IdSportif" : 113, "Nom" : "AVICE" },
{ "IdSportif" : 117, "Nom" : "FERREIRA" },
{ "IdSportif" : 118, "Nom" : "FROMENT" },

```



```

{ "IdSportif" : 124, "Nom" : "SWERTVAEGER" },
{ "IdSportif" : 125, "Nom" : "JALON" },
{ "IdSportif" : 127, "Nom" : "GASTELLIER" },
{ "IdSportif" : 131, "Nom" : "GALLOIS" },
{ "IdSportif" : 133, "Nom" : "LENEVEU" }
],
"count" : 22.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 5 },
  "sportifs" : [
{ "IdSportif" : 74, "Nom" : "AUVRAY" },
{ "IdSportif" : 101, "Nom" : "SWERTVAEGER" },
{ "IdSportif" : 109, "Nom" : "SAUVAGE" },
{ "IdSportif" : 121, "Nom" : "HUET" },
{ "IdSportif" : 126, "Nom" : "DEBOUT" },
{ "IdSportif" : 130, "Nom" : "LECHAUVE" },
{ "IdSportif" : 134, "Nom" : "LERICHE" }
],
"count" : 7.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 4 },
  "sportifs" : [
{ "IdSportif" : 83, "Nom" : "LEROUX" },
{ "IdSportif" : 103, "Nom" : "LEMENOREL" },
{ "IdSportif" : 108, "Nom" : "LAURENCE" },
{ "IdSportif" : 110, "Nom" : "ROULLAND" },
{ "IdSportif" : 135, "Nom" : "MANCEL" }
],
"count" : 5.0
}
{ "_id" : { "Age" : 22, "Conseiller" : 1 },
  "sportifs" : [
{ "IdSportif" : 5, "Nom" : "COMES" },
{ "IdSportif" : 53, "Nom" : "DEMMER" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 3 },
  "sportifs" : [
{ "IdSportif" : 6, "Nom" : "RETALDI" },
{ "IdSportif" : 105, "Nom" : "STILO" },
{ "IdSportif" : 129, "Nom" : "DABON" }
],
"count" : 3.0
}
{ "_id" : { "Age" : 23, "Conseiller" : 7 },
  "sportifs" : [

```

```

{ "IdSportif" : 24, "Nom" : "LEJEUNE" },
{ "IdSportif" : 27, "Nom" : "NIELLEZ" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 24, "Conseiller" : 4 },
  "sportifs" : [
{ "IdSportif" : 30, "Nom" : "SCHINK" },
{ "IdSportif" : 35, "Nom" : "TANQUE" },
{ "IdSportif" : 46, "Nom" : "DAUXIAN" }
],
"count" : 3.0
}
{ "_id" : { "Age" : 26, "Conseiller" : 5 },
  "sportifs" : [
{ "IdSportif" : 141, "Nom" : "BONNET" },
{ "IdSportif" : 143, "Nom" : "BEUZELIN" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 22, "Conseiller" : 5 },
  "sportifs" : [
{ "IdSportif" : 29, "Nom" : "ROUSSEL" },
{ "IdSportif" : 52, "Nom" : "DANDOIS" },
{ "IdSportif" : 92, "Nom" : "HERVIEU" },
{ "IdSportif" : 114, "Nom" : "MICHEL" }
],
"count" : 4.0
}
{ "_id" : { "Age" : 32, "Conseiller" : 3 },
  "sportifs" : [
{ "IdSportif" : 60, "Nom" : "TIZEGHAT" },
{ "IdSportif" : 62, "Nom" : "BAZOUUD" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 24, "Conseiller" : 8 },
  "sportifs" : [
{ "IdSportif" : 12, "Nom" : "BORREL" },
{ "IdSportif" : 16, "Nom" : "COLAS" },
{ "IdSportif" : 39, "Nom" : "MOREL" }
],
"count" : 3.0
}
{ "_id" : { "Age" : 24, "Conseiller" : 1 },
  "sportifs" : [
{ "IdSportif" : 14, "Nom" : "CAILLIOT" },

```

```

{ "IdSportif" : 38, "Nom" : "VASSEMON" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 22, "Conseiller" : 7 },
  "sportifs" : [
{ "IdSportif" : 17, "Nom" : "COLOMB" },
{ "IdSportif" : 41, "Nom" : "DELOVINA" }
],
"count" : 2.0
}
{ "_id" : { "Age" : 30, "Conseiller" : 9 },
  "sportifs" : [
{ "IdSportif" : 70, "Nom" : "BELLAMY" },
{ "IdSportif" : 100, "Nom" : "LESOIF" },
{ "IdSportif" : 107, "Nom" : "BORNE" }
],
"count" : 3.0
}
...

```

s. Quels sportifs n'ont pas de conseillers ?

```
db.Sportifs.find({IdSportifConseiller: null}, {Nom: 1, _id: 0})
```

Résultat (Extrait) :

```
/* 1 */
```

```
{  
  "Nom" : "BOUTAHAR"  
}
```

```
/* 2 */
```

```
{  
  "Nom" : "CHAVANT"  
}
```

```
/* 3 */
```

```
{  
  "Nom" : "CLERICE"  
}
```

```
/* 4 */
```

```
{  
  "Nom" : "CHESNIER"  
}
```

```
/* 5 */
```

```
{  
  "Nom" : "CHAUVIN"  
}
```

```
/* 6 */
```

```
{
```

```

    "Nom" : "RIQUIER"
}

```

```

/* 7 */

```

```

{
    "Nom" : "BONHOMME"
}

```

```

/* 8 */

```

```

{
    "Nom" : "BONE"
}

```

t. Pour chaque sportif donner le nombre de sports qu'il arbitre.

```

db.Sportifs.aggregate([
  { $project: { Nom: 1, Prenom: 1,
    nbSportsArbitres: { $cond: { if: { $isArray: "$Sports.Arbitrer" },
      then: { $size: "$Sports.Arbitrer" }, else: 0 } } } }
])

```

Résultat:

```

/* 1 */

```

```

{
    "_id" : "566eec5f662b388eba464203",
    "Nom" : "BOUTAHAR",
    "Prenom" : "Abderahim",
    "nbSportsArbitres" : 3
}

```

```

/* 2 */

```

```

{
    "_id" : "566eec5f662b388eba464204",
    "Nom" : "KERVADEC",
    "Prenom" : "Yann",
    "nbSportsArbitres" : 2
}

```

```

/* 3 */

```

```

{
    "_id" : "566eec5f662b388eba464205",

```

```

        "Nom" : "HUE",
        "Prenom" : "Pascale",
        "nbSportsArbitres" : 3
    }

/* 4 */
{
    "_id" : "566eec5f662b388eba464206",
    "Nom" : "DORLEANS",
    "Prenom" : "Jean-michel",
    "nbSportsArbitres" : 4
}

/* 5 */
{
    "_id" : "566eec5f662b388eba464207",
    "Nom" : "COMES",
    "Prenom" : "Sylvie",
    "nbSportsArbitres" : 0.0
}

/* 6 */
{
    "_id" : "566eec5f662b388eba464208",
    "Nom" : "RETALDI",
    "Prenom" : "Sophie",
    "nbSportsArbitres" : 3
}

/* 7 */
{
    "_id" : "566eec5f662b388eba464209",
    "Nom" : "GOMEZ",
    "Prenom" : "Diego",
    "nbSportsArbitres" : 3
}

```

```

/* 8 */
{
  "_id" : "566eec5f662b388eba46420a",
  "Nom" : "ANTUNES",
  "Prenom" : "Ga  lle",
  "nbSportsArbitres" : 0.0
}

```

```

/* 9 */
{
  "_id" : "566eec5f662b388eba46420b",
  "Nom" : "BLANDET",
  "Prenom" : "Arnaud",
  "nbSportsArbitres" : 0.0
}

```

```

/* 10 */
{
  "_id" : "566eec5f662b388eba46420c",
  "Nom" : "BONNET",
  "Prenom" : "Christophe",
  "nbSportsArbitres" : 0.0
}

```

```

/* 11 */
{
  "_id" : "566eec5f662b388eba46420d",
  "Nom" : "BORREL",
  "Prenom" : "Beno  t",
  "nbSportsArbitres" : 0.0
}

```

```

/* 12 */
{
  "_id" : "566eec5f662b388eba46420e",
  "Nom" : "BOYON",
  "Prenom" : "Julien",

```

```

        "nbSportsArbitres" : 0.0
    }

/* 13 */
{
    "_id" : "566eec5f662b388eba46420f",
    "Nom" : "CAILLIOT",
    "Prenom" : "Stéphanie",
    "nbSportsArbitres" : 0.0
}

/* 14 */
{
    "_id" : "566eec5f662b388eba464210",
    "Nom" : "CHAMPENOIS",
    "Prenom" : "Cédric",
    "nbSportsArbitres" : 0.0
}

/* 15 */
{
    "_id" : "566eec5f662b388eba464211",
    "Nom" : "COLAS",
    "Prenom" : "Michaël",
    "nbSportsArbitres" : 0.0
}

/* 16 */
{
    "_id" : "566eec5f662b388eba464212",
    "Nom" : "COLOMB",
    "Prenom" : "Michaël",
    "nbSportsArbitres" : 0.0
}

/* 17 */
{

```



```

    "_id" : "566eec5f662b388eba464213",
    "Nom" : "HOSNI",
    "Prenom" : "Leila",
    "nbSportsArbitres" : 0.0
}

```

```

/* 18 */

```

```

{
    "_id" : "566eec5f662b388eba464214",
    "Nom" : "TESTEMONT",
    "Prenom" : "Laurent",
    "nbSportsArbitres" : 0.0
}

```

```

/* 19 */

```

```

{
    "_id" : "566eec5f662b388eba464215",
    "Nom" : "TIENER",
    "Prenom" : "Thomas",
    "nbSportsArbitres" : 0.0
}

```

```

/* 20 */

```

```

{
    "_id" : "566eec5f662b388eba464216",
    "Nom" : "LE BOUCHER",
    "Prenom" : "Denis",
    "nbSportsArbitres" : 0.0
}

```

```

/* 21 */

```

```

{
    "_id" : "566eec5f662b388eba464217",
    "Nom" : "LEDUFAUD",
    "Prenom" : "Pierre",
    "nbSportsArbitres" : 0.0
}

```

```

/* 22 */
{
    "_id" : "566eec5f662b388eba464218",
    "Nom" : "LEGRAND",
    "Prenom" : "David",
    "nbSportsArbitres" : 0.0
}

```

```

/* 23 */
{
    "_id" : "566eec5f662b388eba464219",
    "Nom" : "LEJEUNE",
    "Prenom" : "Sylvaine",
    "nbSportsArbitres" : 0.0
}

```

```

/* 24 */
{
    "_id" : "566eec5f662b388eba46421a",
    "Nom" : "MARTORA",
    "Prenom" : "Fabrice",
    "nbSportsArbitres" : 0.0
}

```

```

/* 25 */
{
    "_id" : "566eec5f662b388eba46421b",
    "Nom" : "MICHEL",
    "Prenom" : "Frédérique",
    "nbSportsArbitres" : 0.0
}

```

```

/* 26 */
{
    "_id" : "566eec5f662b388eba46421c",
    "Nom" : "NIELLEZ",

```

```

        "Prenom" : "Christophe",
        "nbSportsArbitres" : 0.0
    }

/* 27 */
{
    "_id" : "566eec5f662b388eba46421d",
    "Nom" : "RABAHI",
    "Prenom" : "Rabah",
    "nbSportsArbitres" : 0.0
}

/* 28 */
{
    "_id" : "566eec5f662b388eba46421e",
    "Nom" : "ROUSSEL",
    "Prenom" : "Nadège",
    "nbSportsArbitres" : 0.0
}

/* 29 */
{
    "_id" : "566eec5f662b388eba46421f",
    "Nom" : "SCHINK",
    "Prenom" : "Nicolas",
    "nbSportsArbitres" : 0.0
}

/* 30 */
{
    "_id" : "566eec5f662b388eba464220",
    "Nom" : "STEMPUT",
    "Prenom" : "Mathieu",
    "nbSportsArbitres" : 0.0
}

/* 31 */

```

```
{
  "_id" : "566eec5f662b388eba464221",
  "Nom" : "VAN CAUTER",
  "Prenom" : "Vincent",
  "nbSportsArbitres" : 0.0
}
```

```
/* 32 */
```

```
{
  "_id" : "566eec60662b388eba464222",
  "Nom" : "RAMPNOUX",
  "Prenom" : "Jean françois",
  "nbSportsArbitres" : 0.0
}
```

```
/* 33 */
```

```
{
  "_id" : "566eec60662b388eba464223",
  "Nom" : "LHERPINIERE",
  "Prenom" : "Olivier",
  "nbSportsArbitres" : 0.0
}
```

```
/* 34 */
```

```
{
  "_id" : "566eec60662b388eba464224",
  "Nom" : "TANQUE",
  "Prenom" : "Yann",
  "nbSportsArbitres" : 0.0
}
```

```
/* 35 */
```

```
{
  "_id" : "566eec60662b388eba464225",
  "Nom" : "DJELOUDANE",
  "Prenom" : "Zinedine",
  "nbSportsArbitres" : 0.0
}
```

```

}

/* 36 */
{
  "_id" : "566eec60662b388eba464226",
  "Nom" : "LAZZARI",
  "Prenom" : "Magali",
  "nbSportsArbitres" : 0.0
}

/* 37 */
{
  "_id" : "566eec60662b388eba464227",
  "Nom" : "VASSEMON",
  "Prenom" : "Laurent",
  "nbSportsArbitres" : 0.0
}

/* 38 */
{
  "_id" : "566eec60662b388eba464228",
  "Nom" : "MOREL",
  "Prenom" : "Mathieu",
  "nbSportsArbitres" : 0.0
}

/* 39 */
{
  "_id" : "566eec60662b388eba464229",
  "Nom" : "VALIN",
  "Prenom" : "Yann",
  "nbSportsArbitres" : 0.0
}

/* 40 */
{
  "_id" : "566eec60662b388eba46422a",

```

```

        "Nom" : "DELOVINA",
        "Prenom" : "Elina",
        "nbSportsArbitres" : 0.0
    }

/* 41 */
{
    "_id" : "566eec60662b388eba46422b",
    "Nom" : "LEHOUX",
    "Prenom" : "Bruno",
    "nbSportsArbitres" : 0.0
}

/* 42 */
{
    "_id" : "566eec60662b388eba46422c",
    "Nom" : "MATHIEU",
    "Prenom" : "Denise",
    "nbSportsArbitres" : 0.0
}

/* 43 */
{
    "_id" : "566eec60662b388eba46422d",
    "Nom" : "ADIBO",
    "Prenom" : "Senamé",
    "nbSportsArbitres" : 0.0
}

/* 44 */
{
    "_id" : "566eec60662b388eba46422e",
    "Nom" : "CHAVANT",
    "Prenom" : "Christophe",
    "nbSportsArbitres" : 0.0
}

```

```

/* 45 */
{
  "_id" : "566eec60662b388eba46422f",
  "Nom" : "DAUXIAN",
  "Prenom" : "Cédric",
  "nbSportsArbitres" : 0.0
}

```

```

/* 46 */
{
  "_id" : "566eec60662b388eba464230",
  "Nom" : "GROEN",
  "Prenom" : "Céline",
  "nbSportsArbitres" : 0.0
}

```

```

/* 47 */
{
  "_id" : "566eec60662b388eba464231",
  "Nom" : "HEDDI",
  "Prenom" : "Zohra",
  "nbSportsArbitres" : 0.0
}

```

```

/* 48 */
{
  "_id" : "566eec60662b388eba464232",
  "Nom" : "JOUVE",
  "Prenom" : "Sandra",
  "nbSportsArbitres" : 0.0
}

```

```

/* 49 */
{
  "_id" : "566eec60662b388eba464233",
  "Nom" : "KALOMBO",
  "Prenom" : "Yannick",

```

```

        "nbSportsArbitres" : 0.0
    }

/* 50 */

{
    "_id" : "566eec60662b388eba464234",
    "Nom" : "LOPEZ",
    "Prenom" : "Thibaud",
    "nbSportsArbitres" : 0.0
}

```

u. Pour chaque gymnase de Stains donner par jour d'ouverture les horaires des premières et dernières séances.

```

db.Gymnases.aggregate([
  { $match: { Ville: "Stains" } },
  { $unwind: "$Seances" },
  { $group: { _id: { Gymnase: "$NomGymnase", Jour: "$Seances.Jour" },
    premiere_seance: { $min: "$Seances.Horaire" },
    derniere_seance: { $max: "$Seances.Horaire" } } }
])

```

Résultat:

Fetches 0 record(s) in 0ms

v. Pour chaque entraîneur de hand ball quel est le nombre de séances journalières qu'il assure ?

```

db.Gymnases.aggregate([
  { $unwind: "$Seances" },
  { $match: { "Seances.Libelle": "Hand ball" } },
  { $group: { _id: { Entraîneur: "$Seances.IdSportifEntraîneur", Jour:
"$Seances.Jour" },
    nbSeances: { $sum: 1 } } }
])

```

Résultat:

```

/* 1 */

{
    "_id" : {
        "Entraîneur" : 2,
        "Jour" : "mardi"
    },
    "nbSeances" : 1.0
}

/* 2 */

{

```



```

    "_id" : {
      "Entraîneur" : 1,
      "Jour" : "lundi"
    },
    "nbSeances" : 1.0
  }

```

```
/* 3 */
```

```

{
  "_id" : {
    "Entraîneur" : 1,
    "Jour" : "Lundi"
  },
  "nbSeances" : 5.0
}

```

```
/* 4 */
```

```

{
  "_id" : {
    "Entraîneur" : 2,
    "Jour" : "Dimanche"
  },
  "nbSeances" : 2.0
}

```

```
/* 5 */
```

```

{
  "_id" : {
    "Entraîneur" : 3,
    "Jour" : "jeudi"
  },
  "nbSeances" : 1.0
}

```

```
/* 6 */
```

```

{
  "_id" : {

```

```

        "Entraîneur" : 2,
        "Jour" : "mercredi"
    },
    "nbSeances" : 1.0
}

```

```
/* 7 */
```

```

{
    "_id" : {
        "Entraîneur" : 2,
        "Jour" : "Samedi"
    },
    "nbSeances" : 3.0
}

```

```
/* 8 */
```

```

{
    "_id" : {
        "Entraîneur" : 3,
        "Jour" : "dimanche"
    },
    "nbSeances" : 1.0
}

```

```
/* 9 */
```

```

{
    "_id" : {
        "Entraîneur" : 3,
        "Jour" : "mardi"
    },
    "nbSeances" : 1.0
}

```

```
/* 10 */
```

```

{
    "_id" : {
        "Entraîneur" : 3,

```

```

        "Jour" : "lundi"
    },
    "nbSeances" : 4.0
}

```

w. Quels sont les gymnases ayant plus de 15 séances le mercredi ?

```

db.Gymnases.aggregate([
  { $match: { "Seances.Jour": "mercredi" } },
  { $unwind: "$Seances" },
  { $match: { "Seances.Jour": "mercredi" } },
  { $group: { _id: "$NomGymnase", count: { $sum: 1 } } },
  { $match: { count: { $gt: 15 } } }
])

```

Résultat:

Fetches 0 record(s) in 0ms

4. Requêtes avec le paradigme Map-Reduce

a. Calculer le nombre de gymnases pour chaque ville.

```

db.Gymnases.mapReduce(
  function() { emit(this.Ville, 1); },
  function(key, values) { return Array.sum(values); },
  { out: { inline: 1 } }
)

```

Rôle : La fonction map émet la ville comme clé et 1 comme valeur. Le reduce fait la somme des 1 pour compter le nombre de gymnases par ville.

Résultat :

```

/* 1 */
{
  "results" : [
    {
      "_id" : "SAINT DENIS",
      "value" : 3.0
    },
    {
      "_id" : "MONTMORENCY",
      "value" : 5.0
    },
    {
      "_id" : "STAINS",
      "value" : 6.0
    }
  ]
}

```

```

    },
    {
        "_id" : "PIERREFITTE",
        "value" : 5.0
    },
    {
        "_id" : "VILLETANEUSE",
        "value" : 3.0
    },
    {
        "_id" : "GARGES",
        "value" : 1.0
    },
    {
        "_id" : "SARCELLES",
        "value" : 5.0
    }
],
"ok" : 1.0,
"_o" : {
    "results" : [
        {
            "_id" : "SAINT DENIS",
            "value" : 3.0
        },
        {
            "_id" : "MONTMORENCY",
            "value" : 5.0
        },
        {
            "_id" : "STAINS",
            "value" : 6.0
        },
        {
            "_id" : "PIERREFITTE",
            "value" : 5.0
        },
    ],

```

```

        {
            "_id" : "VILLETANEUSE",
            "value" : 3.0
        },
        {
            "_id" : "GARGES",
            "value" : 1.0
        },
        {
            "_id" : "SARCELLES",
            "value" : 5.0
        }
    ],
    "ok" : 1.0
},
"_keys" : [
    "results",
    "ok"
],
"_db" : {
    "_mongo" : {
        "slaveOk" : true,
        "host" : "localhost:27017",
        "defaultDB" : "test",
        "_defaultSession" : {
            "_serverSession" : {
                "client" : {
                    "getReadPreference" : function getReadPreference(driverSession)
{
    "use strict";

```

b. Calculer le nombre de séances pour chaque jour de la semaine.

```

db.Gymnases.mapReduce(
    function() {
        if (this.Seances) {
            this.Seances.forEach(function(seance) {
                if (seance.Jour) {
                    emit(seance.Jour, 1);
                }
            });
        }
    }

```

```

    },
    function(key, values) {
        return Array.sum(values);
    },
    { out: { inline: 1 } }
)

```

Rôle : La fonction map parcourt les séances et émet chaque jour avec la valeur 1. Le reduce somme ces valeurs pour avoir le nombre total de séances par jour.

Résultat:

```

/* 1 */
{
    "results" : [
        {
            "_id" : "Vendredi",
            "value" : 1.0
        },
        {
            "_id" : "Dimanche",
            "value" : 3.0
        },
        {
            "_id" : "Lundi",
            "value" : 9.0
        },
        {
            "_id" : "Samedi",
            "value" : 4.0
        },
        {
            "_id" : "lundi",
            "value" : 14.0
        },
        {
            "_id" : "mardi",
            "value" : 7.0
        },
        {
            "_id" : "mercredi",
            "value" : 12.0
        }
    ]
}

```

```

    },
    {
        "_id" : "jeudi",
        "value" : 5.0
    },
    {
        "_id" : "Mercredi",
        "value" : 1.0
    },
    {
        "_id" : "vendredi",
        "value" : 5.0
    },
    {
        "_id" : "samedi",
        "value" : 5.0
    },
    {
        "_id" : "Mardi",
        "value" : 1.0
    },
    {
        "_id" : "Jeudi",
        "value" : 1.0
    },
    {
        "_id" : "dimanche",
        "value" : 14.0
    }
],
"ok" : 1.0,
"_o" : {
    "results" : [
        {
            "_id" : "Vendredi",
            "value" : 1.0
        },
    ],

```

```

{
  "_id" : "Dimanche",
  "value" : 3.0
},
{
  "_id" : "Lundi",
  "value" : 9.0
},
{
  "_id" : "Samedi",
  "value" : 4.0
},
{
  "_id" : "lundi",
  "value" : 14.0
},
{
  "_id" : "mardi",
  "value" : 7.0
},
{
  "_id" : "mercredi",
  "value" : 12.0
},
{
  "_id" : "jeudi",
  "value" : 5.0
},
{
  "_id" : "Mercredi",
  "value" : 1.0
},
{
  "_id" : "vendredi",
  "value" : 5.0
},
{

```



```

        "_id" : "samedi",
        "value" : 5.0
    },
    {
        "_id" : "Mardi",
        "value" : 1.0
    },
    {
        "_id" : "Jeudi",
        "value" : 1.0
    },
    {
        "_id" : "dimanche",
        "value" : 14.0
    }
],
"ok" : 1.0
},
"_keys" : [
    "results",
    "ok"
],
"_db" : {
    "_mongo" : {
        "slaveOk" : true,
        "host" : "localhost:27017",
        "defaultDB" : "test",
        "_defaultSession" : {
            "_serverSession" : {
                "client" : {
                    "getReadPreference" : function getReadPreference(driverSession)
{
"use strict";

const sessionOptions = driverSession.getOptions();
if (sessionOptions.getReadPreference() !== undefined) {
    return sessionOptions.getReadPreference();

```

```

    }

    return client.getReadPref();
  },

  "getReadConcern" : function getReadConcern(driverSession) {
"use strict";

    const sessionOptions = driverSession.getOptions();
    if (sessionOptions.getReadConcern() !== undefined) {
      return sessionOptions.getReadConcern();
    }
    if (client.getReadConcern() !== undefined) {
      return {level: client.getReadConcern()};
    }
    return null;
  },

  "getWriteConcern" : function getWriteConcern(driverSession) {
"use strict";

    const sessionOptions = driverSession.getOptions();
    if (sessionOptions.getWriteConcern() !== undefined) {
      return sessionOptions.getWriteConcern();
    }
    return client.getWriteConcern();
  },

  "runCommand" : function runCommand(driverSession, dbName,
cmdObj, options) {
"use strict";

    cmdObj = prepareCommandRequest(driverSession, cmdObj);

    const res = runClientFunctionWithRetries(
      driverSession, cmdObj, client.runCommand, [dbName, cmdObj, options]);

    processCommandResponse(driverSession, res);
    return res;
  },

  "runCommandWithMetadata" : function runCommandWithMetadata(

```

```

        driverSession, dbName, metadata, cmdObj) {
"use strict";

        cmdObj = prepareCommandRequest(driverSession, cmdObj);

        const res = runClientFunctionWithRetries(
            driverSession, cmdObj, client.runCommandWithMetadata, [dbName, metadata,
cmdObj]);

        processCommandResponse(driverSession, res);

        return res;
    }

    },

    "handle" : {},

    "isTxnActive" : function isTxnActive() {
"use strict";

        return _txnState === ServerSession.TransactionStates.kActive;
    },

    "isFirstStatement" : function isFirstStatement() {
"use strict";

        return _nextStatementId === 0;
    },

    "getLastUsed" : function getLastUsed() {
"use strict";

        return _lastUsed;
    },

    "getTxnOptions" : function getTxnOptions() {
"use strict";

        return _txnOptions;
    },

    "injectSessionId" : function injectSessionId(cmdObj) {
"use strict";

```

```

cmdObj = Object.assign({}, cmdObj);

const cmdName = Object.keys(cmdObj)[0];

// If the command is in a wrapped form, then we look for the actual command
object

// inside the query/$query object.
let cmdObjUnwrapped = cmdObj;
if (cmdName === "query" || cmdName === "$query") {
    cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);
    cmdObjUnwrapped = cmdObj[cmdName];
}

if (!cmdObjUnwrapped.hasOwnProperty("lsid")) {
    cmdObjUnwrapped.lsid = this.handle.getId();

    // We consider the session to still be in use by the client any time the
session id

    // is injected into the command object as part of making a request.
    updateLastUsed();
}

return cmdObj;
},

    "assignTransactionNumber" : function assignTransactionNumber(cmdObj)
{
    "use strict";

    cmdObj = Object.assign({}, cmdObj);

    const cmdName = Object.keys(cmdObj)[0];

    // If the command is in a wrapped form, then we look for the actual command
object

    // inside the query/$query object.
    let cmdObjUnwrapped = cmdObj;
    if (cmdName === "query" || cmdName === "$query") {
        cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);

```

```

        cmdObjUnwrapped = cmdObj[cmdName];
    }

    if (!cmdObjUnwrapped.hasOwnProperty("txnNumber")) {
        // Since there's no native support for adding NumberLong instances and
        getting back
        // another NumberLong instance, converting from a 64-bit floating-point
        value to a
        // 64-bit integer value will overflow at 2**53.
        _txnNumber++;
        cmdObjUnwrapped.txnNumber = new NumberLong(_txnNumber);
    }

    return cmdObj;
},

    "canRetryWrites" : function canRetryWrites(cmdObj) {
"use strict";

    let cmdName = Object.keys(cmdObj)[0];

    // If the command is in a wrapped form, then we look for the actual command
    name inside
    // the query/$query object.
    if (cmdName === "query" || cmdName === "$query") {
        cmdObj = cmdObj[cmdName];
        cmdName = Object.keys(cmdObj)[0];
    }

    if (isNonNullObject(cmdObj.writeConcern)) {
        const writeConcern = cmdObj.writeConcern;

        // We use bsonWoCompare() in order to handle cases where the "w" field
        is specified
        // as a NumberInt() or NumberLong() instance.
        if (writeConcern.hasOwnProperty("w") &&
            bsonWoCompare({_ : writeConcern.w}, {_ : 0}) === 0) {
            // Unacknowledged writes cannot be retried.
            return false;
        }
    }
}

```

```

    }
}

if (cmdName === "insert") {
    if (!Array.isArray(cmdObj.documents)) {
        // The command object is malformed, so we'll just leave it as-is and
let the
        // server reject it.
        return false;
    }

    // Both single-statement operations (e.g. insertOne()) and multi-
statement
    // operations (e.g. insertMany()) can be retried regardless of whether
they are
    // executed in order by the server.
    return true;
} else if (cmdName === "update") {
    if (!Array.isArray(cmdObj.updates)) {
        // The command object is malformed, so we'll just leave it as-is and
let the
        // server reject it.
        return false;
    }

    const hasMultiUpdate = cmdObj.updates.some(updateOp => updateOp.multi);
    if (hasMultiUpdate) {
        // Operations that modify multiple documents (e.g. updateMany())
cannot be
        // retried.
        return false;
    }

    // Both single-statement operations (e.g. updateOne()) and multi-
statement
    // operations (e.g. bulkWrite()) can be retried regardless of whether
they are
    // executed in order by the server.
    return true;
} else if (cmdName === "delete") {

```

```

        if (!Array.isArray(cmdObj.deletes)) {
            // The command object is malformed, so we'll just leave it as-is and
let the
            // server reject it.
            return false;
        }

        // We use bsonWoCompare() in order to handle cases where the limit is
specified as a
        // NumberInt() or NumberLong() instance.
        const hasMultiDelete = cmdObj.deletes.some(
            deleteOp => bsonWoCompare({_: deleteOp.limit}, {_: 0}) === 0);
        if (hasMultiDelete) {
            // Operations that modify multiple documents (e.g. deleteMany())
cannot be
            // retried.
            return false;
        }

        // Both single-statement operations (e.g. deleteOne()) and multi-
statement
        // operations (e.g. bulkWrite()) can be retried regardless of whether
they are
        // executed in order by the server.
        return true;
    } else if (cmdName === "findAndModify" || cmdName === "findandmodify") {
        // Operations that modify a single document (e.g. findOneAndUpdate())
can be
        // retried.
        return true;
    }

    return false;
},

    "assignTxnInfo" : function assignTxnInfo(cmdObj) {

"use strict";

        // We will want to reset the transaction state to 'inactive' if a normal
operation
        // follows a committed or aborted transaction.

```

```

    if ((_txnState === ServerSession.TransactionStates.kAborted) ||
        (_txnState === ServerSession.TransactionStates.kCommitted &&
            Object.keys(cmdObj)[0] !== "commitTransaction")) {
        _txnState = ServerSession.TransactionStates.kInactive;
    }

    // If we're not in an active transaction or performing a retry on
    commitTransaction,

    // return early.
    if (_txnState === ServerSession.TransactionStates.kInactive) {
        return cmdObj;
    }

    // If we reconnect to a 3.6 server in the middle of a transaction, we
    // catch it here.
    if (!serverSupports(kWireVersionSupportingMultiDocumentTransactions)) {
        _txnState = ServerSession.TransactionStates.kInactive;
        throw new Error(
            "Transactions are only supported on server versions 4.0 and
greater.");
    }

    cmdObj = Object.assign({}, cmdObj);

    const cmdName = Object.keys(cmdObj)[0];

    // If the command is in a wrapped form, then we look for the actual command
    object

    // inside the query/$query object.
    let cmdObjUnwrapped = cmdObj;
    if (cmdName === "query" || cmdName === "$query") {
        cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);
        cmdObjUnwrapped = cmdObj[cmdName];
    }

    if (!cmdObjUnwrapped.hasOwnProperty("txnNumber")) {
        // Since there's no native support for adding NumberLong instances and
        getting back

        // another NumberLong instance, converting from a 64-bit floating-point
        value to a

```



```

        // 64-bit integer value will overflow at 2**53.
        cmdObjUnwrapped.txnNumber = new NumberLong(_txnNumber);
    }

    // All operations of a multi-statement transaction must specify
    autocommit=false.

    cmdObjUnwrapped.autocommit = false;

    // Statement Id is required on all transaction operations.
    cmdObjUnwrapped.stmtId = new NumberInt(_nextStatementId);

    // 'readConcern' and 'startTransaction' can only be specified on the first
statement
    // in a transaction.
    if (_nextStatementId == 0) {
        cmdObjUnwrapped.startTransaction = true;
        if (_txnOptions.getTxnReadConcern() !== undefined) {
            // Override the readConcern with the one specified during
startTransaction.
            cmdObjUnwrapped.readConcern = Object.assign(
                {}, cmdObjUnwrapped.readConcern,
                _txnOptions.getTxnReadConcern());
        }
    }

    // Reserve the statement ids for batch writes.
    switch (cmdName) {
        case "insert":
            _nextStatementId += cmdObjUnwrapped.documents.length;
            break;
        case "update":
            _nextStatementId += cmdObjUnwrapped.updates.length;
            break;
        case "delete":
            _nextStatementId += cmdObjUnwrapped.deletes.length;
            break;
        default:
            _nextStatementId += 1;
    }

```

```

    }

    return cmdObj;
  },

  "startTransaction" : function startTransaction(txnOptsObj) {
"use strict";

    // If the session is already in a transaction, raise an error.
    if (this.isTxnActive()) {
        throw new Error("Transaction already in progress on this session.");
    }

    if (!serverSupports(kWireVersionSupportingMultiDocumentTransactions)) {
        throw new Error(
            "Transactions are only supported on server versions 4.0 and
greater.");
    }

    _txnOptions = new TransactionOptions(txnOptsObj);
    _txnState = ServerSession.TransactionStates.kActive;
    _nextStatementId = 0;
    _txnNumber++;
  },

  "commitTransaction" : function commitTransaction(driverSession) {
"use strict";

    // If the transaction state is already 'aborted' we cannot try to commit it.
    if (_txnState === ServerSession.TransactionStates.kAborted) {
        throw new Error("Cannot call commitTransaction after calling
abortTransaction.");
    }

    // If the session has no active transaction, raise an error.
    if (_txnState === ServerSession.TransactionStates.kInactive) {
        throw new Error("There is no active transaction to commit on this
session.");
    }

    // run commitTxn command
    return endTransaction("commitTransaction", driverSession);
  },

  "abortTransaction" : function abortTransaction(driverSession) {

```

```

"use strict";

        // If the transaction state is already 'aborted' we cannot try to abort it
again.
        if (_txnState === ServerSession.TransactionStates.kAborted) {
            throw new Error("Cannot call abortTransaction twice.");
        }
        // We cannot attempt to abort a transaction that has already been committed.
        if (_txnState === ServerSession.TransactionStates.kCommitted) {
            throw new Error("Cannot call abortTransaction after calling
commitTransaction.");
        }
        // If the session has no active transaction, raise an error.
        if (_txnState === ServerSession.TransactionStates.kInactive) {
            throw new Error("There is no active transaction to abort on this
session.");
        }
        // run abortTxn command
        return endTransaction("abortTransaction", driverSession);
    }

    },
    "_isExplicit" : false,
    "getClient" : function getClient() {
        "use strict";

        return client;
    },
    "_getSessionAwareClient" : function _getSessionAwareClient() {
        "use strict";

        return this._serverSession.client;
    },
    "getOptions" : function getOptions() {
        "use strict";

        return _options;
    },
    "getSessionId" : function getSessionId() {

```

```

"use strict";

    if (!this._serverSession.hasOwnProperty("handle")) {
        return null;
    }

    return this._serverSession.handle.getId();
},

"getOperationTime" : function getOperationTime() {
"use strict";

    return _operationTime;
},

"advanceOperationTime" : function advanceOperationTime(operationTime) {
"use strict";

    if (!isNonNullObject(_operationTime) ||
        bsonWoCompare({_ : operationTime}, {_ : _operationTime}) > 0) {
        _operationTime = operationTime;
    }
},

"resetOperationTime_forTesting" : function
resetOperationTime_forTesting() {
"use strict";

    _operationTime = undefined;
},

"getClusterTime" : function getClusterTime() {
"use strict";

    return _clusterTime;
},

"advanceClusterTime" : function advanceClusterTime(clusterTime) {
"use strict";

    if (!isNonNullObject(_clusterTime) ||
        bsonWoCompare({_ : clusterTime.clusterTime}, {_ :
_clusterTime.clusterTime}) >

```

```

        0) {
            _clusterTime = clusterTime;
        }
    },
    "resetClusterTime_forTesting" : function resetClusterTime_forTesting() {
"use strict";

        _clusterTime = undefined;
    },
    "getDatabase" : function getDatabase(dbName) {
"use strict";

        const db = client.getDB(dbName);
        db._session = this;
        return db;
    },
    "hasEnded" : function hasEnded() {
"use strict";

        return _hasEnded;
    },
    "endSession" : function endSession() {
"use strict";

        if (this._hasEnded) {
            return;
        }

        this._hasEnded = true;
        implMethods.endSession(this._serverSession);
    },
    "shellPrint" : function () {
"use strict";

        return this.toString();
    },
    "tojson" : function _tojson(...args) {

```

```

"use strict";

        return tojson(this.getSessionId(), ...args);
    },
    "toString" : function toString() {
        "use strict";

        const sessionId = this.getSessionId();
        if (sessionId === null) {
            return "dummy session";
        }
        return "session " + tojson(sessionId);
    },
    "startTransaction" : function startTransaction(txnOptsObj = {}) {
        "use strict";

        this._serverSession.startTransaction(txnOptsObj);
    },
    "commitTransaction" : function commitTransaction() {
        "use strict";

        assert.commandWorked(this._serverSession.commitTransaction(this));
    },
    "abortTransaction" : function abortTransaction() {
        "use strict";

        // Intentionally ignore command result.
        this._serverSession.abortTransaction(this);
    },
    "commitTransaction_forTesting" : function commitTransaction_forTesting()
{
    "use strict";

    return this._serverSession.commitTransaction(this);
},
    "abortTransaction_forTesting" : function abortTransaction_forTesting() {
        "use strict";

```

```

        return this._serverSession.abortTransaction(this);
    }
},
    "_readMode" : "commands"
},
    "_name" : "gym",
    "_session" : {
        "_serverSession" : {
            "client" : {
                "getReadPreference" : function getReadPreference(driverSession) {
"use strict";

                const sessionOptions = driverSession.getOptions();
                if (sessionOptions.getReadPreference() !== undefined) {
                    return sessionOptions.getReadPreference();
                }
                return client.getReadPref();
            },
            "getReadConcern" : function getReadConcern(driverSession) {
"use strict";

                const sessionOptions = driverSession.getOptions();
                if (sessionOptions.getReadConcern() !== undefined) {
                    return sessionOptions.getReadConcern();
                }
                if (client.getReadConcern() !== undefined) {
                    return {level: client.getReadConcern()};
                }
                return null;
            },
            "getWriteConcern" : function getWriteConcern(driverSession) {
"use strict";

                const sessionOptions = driverSession.getOptions();
                if (sessionOptions.getWriteConcern() !== undefined) {
                    return sessionOptions.getWriteConcern();

```

```

    }
    return client.getWriteConcern();
  },
  "runCommand" : function runCommand(driverSession, dbName, cmdObj,
options) {
  "use strict";

  cmdObj = prepareCommandRequest(driverSession, cmdObj);

  const res = runClientFunctionWithRetries(
    driverSession, cmdObj, client.runCommand, [dbName, cmdObj, options]);

  processCommandResponse(driverSession, res);
  return res;
},
  "runCommandWithMetadata" : function runCommandWithMetadata(
driverSession, dbName, metadata, cmdObj) {
  "use strict";

  cmdObj = prepareCommandRequest(driverSession, cmdObj);

  const res = runClientFunctionWithRetries(
    driverSession, cmdObj, client.runCommandWithMetadata, [dbName, metadata,
cmdObj]);

  processCommandResponse(driverSession, res);
  return res;
}
},
  "handle" : {},
  "isTxnActive" : function isTxnActive() {
  "use strict";

  return _txnState === ServerSession.TransactionStates.kActive;
},
  "isFirstStatement" : function isFirstStatement() {
  "use strict";

```



```

        return _nextStatementId === 0;
    },
    "getLastUsed" : function getLastUsed() {
"use strict";

        return _lastUsed;
    },
    "getTxnOptions" : function getTxnOptions() {
"use strict";

        return _txnOptions;
    },
    "injectSessionId" : function injectSessionId(cmdObj) {
"use strict";

        cmdObj = Object.assign({}, cmdObj);

        const cmdName = Object.keys(cmdObj)[0];

object    // If the command is in a wrapped form, then we look for the actual command

        // inside the query/$query object.
        let cmdObjUnwrapped = cmdObj;
        if (cmdName === "query" || cmdName === "$query") {
            cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);
            cmdObjUnwrapped = cmdObj[cmdName];
        }

        if (!cmdObjUnwrapped.hasOwnProperty("lsid")) {
            cmdObjUnwrapped.lsid = this.handle.getId();

session id    // We consider the session to still be in use by the client any time the

        // is injected into the command object as part of making a request.
        updateLastUsed();
    }

```

```

        return cmdObj;
    },

    "assignTransactionNumber" : function assignTransactionNumber(cmdObj) {
"use strict";

        cmdObj = Object.assign({}, cmdObj);

        const cmdName = Object.keys(cmdObj)[0];

        // If the command is in a wrapped form, then we look for the actual command
object
        // inside the query/$query object.
        let cmdObjUnwrapped = cmdObj;
        if (cmdName === "query" || cmdName === "$query") {
            cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);
            cmdObjUnwrapped = cmdObj[cmdName];
        }

        if (!cmdObjUnwrapped.hasOwnProperty("txnNumber")) {
            // Since there's no native support for adding NumberLong instances and
getting back
            // another NumberLong instance, converting from a 64-bit floating-point
value to a
            // 64-bit integer value will overflow at 2**53.
            _txnNumber++;
            cmdObjUnwrapped.txnNumber = new NumberLong(_txnNumber);
        }

        return cmdObj;
    },

    "canRetryWrites" : function canRetryWrites(cmdObj) {
"use strict";

        let cmdName = Object.keys(cmdObj)[0];

        // If the command is in a wrapped form, then we look for the actual command
name inside

```

```

// the query/$query object.
if (cmdName === "query" || cmdName === "$query") {
    cmdObj = cmdObj[cmdName];
    cmdName = Object.keys(cmdObj)[0];
}

if (isNonNullObject(cmdObj.writeConcern)) {
    const writeConcern = cmdObj.writeConcern;

    // We use bsonWoCompare() in order to handle cases where the "w" field
is specified

    // as a NumberInt() or NumberLong() instance.
    if (writeConcern.hasOwnProperty("w") &&
        bsonWoCompare({_: writeConcern.w}, {_: 0}) === 0) {
        // Unacknowledged writes cannot be retried.
        return false;
    }
}

if (cmdName === "insert") {
    if (!Array.isArray(cmdObj.documents)) {
        // The command object is malformed, so we'll just leave it as-is and
let the

        // server reject it.
        return false;
    }

    // Both single-statement operations (e.g. insertOne()) and multi-
statement

    // operations (e.g. insertMany()) can be retried regardless of whether
they are

    // executed in order by the server.
    return true;
} else if (cmdName === "update") {
    if (!Array.isArray(cmdObj.updates)) {
        // The command object is malformed, so we'll just leave it as-is and
let the

        // server reject it.
        return false;
    }
}

```

```

    }

    const hasMultiUpdate = cmdObj.updates.some(updateOp => updateOp.multi);
    if (hasMultiUpdate) {
        // Operations that modify multiple documents (e.g. updateMany())
        // retried.
        return false;
    }

    // Both single-statement operations (e.g. updateOne()) and multi-
    // operations (e.g. bulkWrite()) can be retried regardless of whether
    // executed in order by the server.
    return true;
} else if (cmdName === "delete") {
    if (!Array.isArray(cmdObj.deletes)) {
        // The command object is malformed, so we'll just leave it as-is and
        // server reject it.
        return false;
    }

    // We use bsonWoCompare() in order to handle cases where the limit is
    // NumberInt() or NumberLong() instance.
    const hasMultiDelete = cmdObj.deletes.some(
        deleteOp => bsonWoCompare({_: deleteOp.limit}, {_: 0}) === 0);
    if (hasMultiDelete) {
        // Operations that modify multiple documents (e.g. deleteMany())
        // retried.
        return false;
    }

    // Both single-statement operations (e.g. deleteOne()) and multi-
    // operations (e.g. bulkWrite()) can be retried regardless of whether

```

```

        // executed in order by the server.
        return true;
    } else if (cmdName === "findAndModify" || cmdName === "findandmodify") {
        // Operations that modify a single document (e.g. findOneAndUpdate())
        // retried.
        return true;
    }

    return false;
},

    "assignTxnInfo" : function assignTxnInfo(cmdObj) {
"use strict";

        // We will want to reset the transaction state to 'inactive' if a normal
operation
        // follows a committed or aborted transaction.
        if ((_txnState === ServerSession.TransactionStates.kAborted) ||
            (_txnState === ServerSession.TransactionStates.kCommitted &&
            Object.keys(cmdObj)[0] !== "commitTransaction")) {
            _txnState = ServerSession.TransactionStates.kInactive;
        }

        // If we're not in an active transaction or performing a retry on
commitTransaction,
        // return early.
        if (_txnState === ServerSession.TransactionStates.kInactive) {
            return cmdObj;
        }

        // If we reconnect to a 3.6 server in the middle of a transaction, we
        // catch it here.
        if (!serverSupports(kWireVersionSupportingMultiDocumentTransactions)) {
            _txnState = ServerSession.TransactionStates.kInactive;
            throw new Error(
greater.");
        }
    }
}

```

```

cmdObj = Object.assign({}, cmdObj);

const cmdName = Object.keys(cmdObj)[0];

// If the command is in a wrapped form, then we look for the actual command
object
// inside the query/$query object.
let cmdObjUnwrapped = cmdObj;
if (cmdName === "query" || cmdName === "$query") {
    cmdObj[cmdName] = Object.assign({}, cmdObj[cmdName]);
    cmdObjUnwrapped = cmdObj[cmdName];
}

if (!cmdObjUnwrapped.hasOwnProperty("txnNumber")) {
    // Since there's no native support for adding NumberLong instances and
    getting back
    // another NumberLong instance, converting from a 64-bit floating-point
    value to a
    // 64-bit integer value will overflow at 2**53.
    cmdObjUnwrapped.txnNumber = new NumberLong(_txnNumber);
}

// All operations of a multi-statement transaction must specify
autocommit=false.
cmdObjUnwrapped.autocommit = false;

// Statement Id is required on all transaction operations.
cmdObjUnwrapped.stmtId = new NumberInt(_nextStatementId);

// 'readConcern' and 'startTransaction' can only be specified on the first
statement
// in a transaction.
if (_nextStatementId == 0) {
    cmdObjUnwrapped.startTransaction = true;
    if (_txnOptions.getTxnReadConcern() !== undefined) {
        // Override the readConcern with the one specified during
        startTransaction.
        cmdObjUnwrapped.readConcern = Object.assign(
            {}, cmdObjUnwrapped.readConcern,
            _txnOptions.getTxnReadConcern());
    }
}

```

```

        }
    }

    // Reserve the statement ids for batch writes.
    switch (cmdName) {
        case "insert":
            _nextStatementId += cmdObjUnwrapped.documents.length;
            break;
        case "update":
            _nextStatementId += cmdObjUnwrapped.updates.length;
            break;
        case "delete":
            _nextStatementId += cmdObjUnwrapped.deletes.length;
            break;
        default:
            _nextStatementId += 1;
    }

    return cmdObj;
},

    "startTransaction" : function startTransaction(txnOptsObj) {
"use strict";

    // If the session is already in a transaction, raise an error.
    if (this.isTxnActive()) {
        throw new Error("Transaction already in progress on this session.");
    }

    if (!serverSupports(kWireVersionSupportingMultiDocumentTransactions)) {
        throw new Error(
            "Transactions are only supported on server versions 4.0 and
greater.");
    }

    _txnOptions = new TransactionOptions(txnOptsObj);
    _txnState = ServerSession.TransactionStates.kActive;
    _nextStatementId = 0;
    _txnNumber++;
},

```

```

        "commitTransaction" : function commitTransaction(driverSession) {
"use strict";

        // If the transaction state is already 'aborted' we cannot try to commit it.
        if (_txnState === ServerSession.TransactionStates.kAborted) {
            throw new Error("Cannot call commitTransaction after calling
abortTransaction.");
        }

        // If the session has no active transaction, raise an error.
        if (_txnState === ServerSession.TransactionStates.kInactive) {
            throw new Error("There is no active transaction to commit on this
session.");
        }

        // run commitTxn command
        return endTransaction("commitTransaction", driverSession);
    },

    "abortTransaction" : function abortTransaction(driverSession) {
"use strict";

        // If the transaction state is already 'aborted' we cannot try to abort it
again.
        if (_txnState === ServerSession.TransactionStates.kAborted) {
            throw new Error("Cannot call abortTransaction twice.");
        }

        // We cannot attempt to abort a transaction that has already been committed.
        if (_txnState === ServerSession.TransactionStates.kCommitted) {
            throw new Error("Cannot call abortTransaction after calling
commitTransaction.");
        }

        // If the session has no active transaction, raise an error.
        if (_txnState === ServerSession.TransactionStates.kInactive) {
            throw new Error("There is no active transaction to abort on this
session.");
        }

        // run abortTxn command
        return endTransaction("abortTransaction", driverSession);
    }
},

    "_isExplicit" : false,

```



```

        "getClient" : function getClient() {
"use strict";

            return client;
        },
        "_getSessionAwareClient" : function _getSessionAwareClient() {
"use strict";

            return this._serverSession.client;
        },
        "getOptions" : function getOptions() {
"use strict";

            return _options;
        },
        "getSessionId" : function getSessionId() {
"use strict";

            if (!this._serverSession.hasOwnProperty("handle")) {
                return null;
            }
            return this._serverSession.handle.getId();
        },
        "getOperationTime" : function getOperationTime() {
"use strict";

            return _operationTime;
        },
        "advanceOperationTime" : function advanceOperationTime(operationTime) {
"use strict";

            if (!isNonNullObject(_operationTime) ||
                bsonWoCompare({_ : operationTime}, {_ : _operationTime}) > 0) {
                _operationTime = operationTime;
            }
        },
        "resetOperationTime_forTesting" : function resetOperationTime_forTesting() {

```

```

"use strict";

        _operationTime = undefined;
    },
    "getClusterTime" : function getClusterTime() {
"use strict";

        return _clusterTime;
    },
    "advanceClusterTime" : function advanceClusterTime(clusterTime) {
"use strict";

        if (!isNonNullObject(_clusterTime) ||
            bsonWoCompare({_ : clusterTime.clusterTime}, {_ :
_clusterTime.clusterTime}) >
                0) {
            _clusterTime = clusterTime;
        }
    },
    "resetClusterTime_forTesting" : function resetClusterTime_forTesting() {
"use strict";

        _clusterTime = undefined;
    },
    "getDatabase" : function getDatabase(dbName) {
"use strict";

        const db = client.getDB(dbName);

        db._session = this;

        return db;
    },
    "hasEnded" : function hasEnded() {
"use strict";

        return _hasEnded;
    },
    "endSession" : function endSession() {

```

```

"use strict";

        if (this._hasEnded) {
            return;
        }

        this._hasEnded = true;
        implMethods.endSession(this._serverSession);
    },
    "shellPrint" : function () {
"use strict";

        return this.toString();
    },
    "tojson" : function _tojson(...args) {
"use strict";

        return tojson(this.getSessionId(), ...args);
    },
    "toString" : function toString() {
"use strict";

        const sessionId = this.getSessionId();
        if (sessionId === null) {
            return "dummy session";
        }
        return "session " + tojson(sessionId);
    },
    "startTransaction" : function startTransaction(txnOptsObj = {}) {
"use strict";

        this._serverSession.startTransaction(txnOptsObj);
    },
    "commitTransaction" : function commitTransaction() {
"use strict";

        assert.commandWorked(this._serverSession.commitTransaction(this));

```



```

{ "_id": "Hockey", "value": 27.0 }
],
"ok": 1.0,
"_o": {
  "results": [
{ "_id": "Volley ball", "value": 28.0 },
{ "_id": "Hand ball", "value": 20.0 },
{ "_id": "Basket ball", "value": 7.0 },
{ "_id": "Hockey", "value": 27.0 }
],
"ok": 1.0
},
"_keys": [
  "results",
  "ok"
],
"_db": {
  "_mongo": {
    "slaveOk": true,
    "host": "localhost:27017",
    "defaultDB": "test",
    "_defaultSession": {
      "_serverSession": {
        "client": {
          "getReadPreference": function getReadPreference(driverSession) {
"use strict";
...

```

d. Calculer la superficie moyenne des gymnases, pour chaque ville.

```

db.Gymnases.mapReduce (
  // MAP : émettre {somme, count} par ville
  function() {
    if (this.Ville && this.Surface) {
      emit(this.Ville, { somme: this.Surface, count: 1 });
    }
  },
  // REDUCE : additionner les sommes ET les counts
  function(key, values) {
    var result = { somme: 0, count: 0 };
    values.forEach(function(v) {
      result.somme += v.somme;
      result.count += v.count;
    });
    return result;
  },
  // FINALIZE : calculer la moyenne à partir de somme/count
  {
    out: { inline: 1 },
    finalize: function(key, reduced) {

```

```

        reduced.moyenne = reduced.somme / reduced.count;
        return reduced;
    }
}
)

```

Rôle : La fonction map émet la ville avec un objet contenant la surface et 1. Le reduce fait la somme. Le finalize calcule la moyenne finale (surface/nombre).

Résultat :

```
````javascript
```

```

{ "results" : [{ "_id" : "SAINT DENIS", "value" : { "somme" : 1470.0, "count" : 3.0, "moyenne" :
490.0 }
},
{ "_id" : "PIERREFITTE", "value" : { "somme" : 1910.0, "count" : 5.0, "moyenne" : 382.0 }
},
{ "_id" : "STAINS", "value" : { "somme" : 2000.0, "count" : 6.0, "moyenne" : 333.3333333333333 }
},
{ "_id" : "MONTMORENCY", "value" : { "somme" : 2350.0, "count" : 5.0, "moyenne" : 470.0 }
},
{ "_id" : "VILLETANEUSE", "value" : { "somme" : 1570.0, "count" : 3.0, "moyenne" :
523.3333333333333 }
},
{ "_id" : "GARGES", "value" : { "somme" : 400.0, "count" : 1.0, "moyenne" : 400.0 }
},
{ "_id" : "SARCELLES", "value" : { "somme" : 2740.0, "count" : 5.0, "moyenne" : 548.0 }
}
],
"ok" : 1.0,
"_o" : {
 "results" : [
{ "_id" : "SAINT DENIS", "value" : { "somme" : 1470.0, "count" : 3.0, "moyenne" : 490.0 }
},
{ "_id" : "PIERREFITTE", "value" : { "somme" : 1910.0, "count" : 5.0, "moyenne" : 382.0 }
},
{ "_id" : "STAINS", "value" : { "somme" : 2000.0, "count" : 6.0, "moyenne" : 333.3333333333333 }
},
{ "_id" : "MONTMORENCY", "value" : { "somme" : 2350.0, "count" : 5.0, "moyenne" : 470.0 }
},
{ "_id" : "VILLETANEUSE", "value" : { "somme" : 1570.0, "count" : 3.0, "moyenne" :
523.3333333333333 }
},
{ "_id" : "GARGES", "value" : { "somme" : 400.0, "count" : 1.0, "moyenne" : 400.0 }
},
{ "_id" : "SARCELLES", "value" : { "somme" : 2740.0, "count" : 5.0, "moyenne" : 548.0 }
}
],
"ok" : 1.0
},

```

```

 "_keys" : [
 "results",
 "ok"
],
 "_db" : {
 "_mongo" : {
 "slaveOk" : true,
 "host" : "localhost:27017",
 "defaultDB" : "test",
 "_defaultSession" : {
 "_serverSession" : {
 "client" : {
 "getReadPreference" : function getReadPreference(driverSession) {
"use strict";

```

#### Conclusion :

Ce TP a permis d'assimiler les concepts fondamentaux de MongoDB, la manipulation de collections et documents avec le format JSON/BSON, ainsi que l'utilisation du framework d'agrégation et du paradigme Map-Reduce pour des calculs statistiques avancés. L'ensemble de la base de données travaillée a été exporté sous format JSON pour archivage.

'''